Proof of Useful Quantum Work (PoUQW)

Version 1.0

Abstract

We present **Proof of Useful Quantum Work (PoUQW)**, a Layer 1 blockchain protocol with a novel consensus primitive based on economically productive hybrid quantum-classical computations. In PoUQW, block production requires miners to complete parameterized quantum computational tasks — such as variational quantum eigensolving for molecular ground-state estimation, quantum approximate optimization for combinatorial problems, or quantum kernel evaluation for machine learning — drawn from a governance-curated on-chain Task Registry. Each task exploits a fundamental asymmetry: the quantum computation is classically hard to *perform* but classically efficient to *verify*, providing an NP-like structure that enables any classical node to validate work proofs without access to quantum hardware.

Validator selection is driven by a **quantum random beacon** built on self-certifying quantum random number generators (SC-QRNGs) — that transform a physical quantum entropy source into a continuous, mathematically rigorous min-entropy certificate for every output batch. Unlike classical RANDAO (where randomness generation is grounded on economic game theory) or blind-trust QRNGs (where the user blindly trusts the hardware manufacturer), SC-QRNGs provide physics-grounded, continuously auditable randomness. Multiple nodes aggregate their outputs via XOR, ensuring that a single certified node's randomness suffices for unbiased validator selection.

The protocol exposes a fully **EVM-compatible execution layer**, enabling general-purpose smart contracts written in Solidity to execute on the chain using standard Ethereum tooling. Special precompiled contracts allow smart contracts to natively consume quantum computation results and certified random values, bridging the quantum work layer with decentralized applications.

All mechanisms described in this paper are designed for **near-term noisy intermediate-scale quantum (NISQ) hardware** with 50–1000+ physical qubits, as available from current vendors including IBM (Eagle/Heron processors), Quantinuum (H-series trapped-ion systems), and IonQ (Forte). We do not invoke fault-tolerant quantum computing, logical qubits, or speculative future hardware capabilities. The protocol's security against quantum attacks on its own cryptographic primitives is ensured through post-quantum lattice-based signatures (CRYSTALS-Dilithium) and hash-based commitments.

We analyze the security properties of PoUQW, including resistance to proof fabrication, Sybil attacks, randomness manipulation, and majority-power adversaries.

Contents

1 Introduction **4**

1.1 The Consensus Trilemma of Useful Work . . . 4

1.2 The Three Pillars of PoUQW . . . 5

1.3 Design Philosophy and Scope . . . 5

2 Background and Related Work **6**

2.1 Proof-of-Work and Its Discontents . . . 6

2.2 Proof-of-Stake and Economic Consensus . . . 6

2.3 Hybrid Quantum-Classical Algorithms and the Verifiability Gap . . . 7

2.4 Quantum Random Number Generation . . . 8

2.5 Quantum-Resistant Versus Quantum-Native Blockchains . . . 8

2.6 Bittensor and Proof of Useful Work for AI . . . 9

2.6.1 Bittensor's Design . . . 9

2.6.2 What Bittensor Gets Right . . . 9

2.6.3 Where PoUQW Diverges . . . 10

2.6.4 Philosophical Framing . . . 11

3 System Architecture **12**

3.1 High-Level Overview . . . 12

3.2 Layer Descriptions . . . 12

3.3 Inter-Layer Communication . . . 14

3.4 Node Types . . . 14

4 Proof of Useful Quantum Work — Detailed Design **14**

4.1 Eligible Work Tasks: The Task Registry . . . 15

4.1.1 Molecular Ground-State Energy (VQE) . . . 15

4.1.2 Combinatorial Optimization (QAOA) . . . 17

4.1.3 Quantum Chemistry Excited States (VQD) . . . 18

4.1.4 Linear Systems (VQLS) . . . 19

4.1.5 Quantum Machine Learning Feature Maps (Quantum Kernel Estimation) 20

4.2 Work Proof Format . . . 20

4.3 Difficulty Adjustment . . . 21

5 Quantum Random Beacon for Validator Selection **21**

5.1 Self-Certifying Quantum Randomness . . . 21

5.1.1 Self-Certifying QRNGs . . . 22

5.2 Certification and Externally Verifiable Security . . . 23

5.2.1 Randomness Extraction . . . 23

5.2.2 Security Assumptions . . . 24

5.3 Deployment Options . . . 26

5.4 Beacon Protocol . . . 26

5.4.1 Epoch Structure . . . 27

5.4.2 Comparison to existing approaches for randomness generation 27
5.5 Validator Election 28
5.5.1 Staked Validator Set 28
5.5.2 Deterministic Selection from Beacon Output 28

6 Consensus Protocol **29**
6.1 Block Production 29
6.2 BFT Finality 29
6.3 Fork Choice Rule 29
6.4 Incentives and Tokenomics Overview 29
6.4.1 Block Rewards 29
6.4.2 Staking Rewards and Slashing 30
6.4.3 Task Bounty Market 30
6.4.4 EVM Execution Layer 31

7 Use Cases **31**
7.1 Drug Discovery as Mining 31
7.2 Financial Portfolio Optimization 32
7.3 Provably Fair Randomness for Gaming and Lotteries 32
7.4 Supply Chain and Logistics Optimization 33
7.5 Quantum Machine Learning 33
7.6 Materials Science 34
7.7 Nuclear Structure Simulation 34
7.8 Catalyst Design for Clean Energy 35

8 Token Economics: A Store of Value for Quantum Computing **35**
8.1 The Commodity Thesis 35
8.2 The Quantum Computing Reserve Asset Thesis 37

9 Security Analysis **38**
9.1 Sybil Resistance 38
9.2 Proof Fabrication 39
9.3 Randomness Manipulation 40
9.4 Majority Power Attack 40
9.5 Quantum Threats to Cryptography 41

10 The PoUQW Backbone Protocol **42**
10.1 Notation 42
10.2 Algorithm 1: Per-Slot Node Protocol 42
10.3 Algorithm 2: Classical Verification 43
10.4 Protocol Lifecycle 44

1 Introduction

1.1 The Consensus Trilemma of Useful Work

The foundational innovation of Bitcoin was the demonstration that a decentralized network of mutually distrustful parties may reach consensus on a shared ledger through Proof-of-Work (PoW): the expenditure of computational resources on a cryptographic hash puzzle whose solution is trivial to verify but expensive to find [25]. This asymmetry — hard to produce, easy to check — is the structural engine of trustless consensus. Yet Bitcoin's PoW carries a well-documented externality: the computational work itself is intentionally *useless*. The SHA-256 partial preimage searches that secure the Bitcoin network produce no output of independent economic or scientific value. As of 2026, the Bitcoin network consumes approximately 150 TWh of electricity annually — comparable to the energy consumption of a mid-sized nation — entirely on hash puzzles whose outputs are discarded after each block [34].

Proof-of-Stake (PoS), as implemented in Ethereum's Beacon Chain since September 2022, eliminates this energy waste by replacing computational work with economic collateral. Validators lock capital (staked ETH) and are selected to propose and attest to blocks in proportion to their stake. The security model shifts from thermodynamic cost to economic cost: attacking the network requires acquiring and risking a majority of staked capital [7]. However, PoS introduces its own structural concern — the *concentration of consensus power in capital holders*. Validator influence scales with wealth, not with productive contribution, creating a plutocratic dynamic where the rich get richer through compounding staking yields. Moreover, the work performed by validators — block proposal and attestation — is computationally trivial and produces no independent economic output.

A natural question emerges: **can consensus security be bootstrapped from computations that are simultaneously (a) hard to perform, (b) easy to verify, and (c) economically or scientifically useful?**

Several projects have explored fragments of this question. Primecoin [19] directed PoW toward finding Cunningham chains of prime numbers — a mathematical curiosity with limited commercial value. Chia [8] replaced hash mining with Proofs of Space and Time based on Verifiable Delay Functions, reducing energy consumption but producing no useful computational output. Various proposals for AI-based Proof-of-Useful-Work, most notably Bittensor [29], have attempted to channel consensus computation toward machine learning training and inference — with significant traction but fundamental challenges in verification objectivity, which we analyze in Section 2.

This paper presents **Proof of Useful Quantum Work (PoUQW)**, a protocol that resolves the useful-work consensus problem by exploiting a structural property of hybrid quantum-classical computation: **variational quantum algorithms produce outputs that are classically hard to find but classically efficient to verify**. A Variational Quantum Eigensolver (VQE) circuit, for instance, prepares trial states in an exponentially large Hilbert space and uses a classical optimizer to search over the parameter space $\theta \in \mathbb{R}^p$ to find θ^* that minimizes the expectation value $\langle\psi(\theta^*)|H|\psi(\theta^*)\rangle$ of a molecular Hamiltonian H. This search is conjectured to be intractable for classical computers at sufficient scale. Yet *verifying* the result —

computing $\langle\psi(\theta^*)|H|\psi(\theta^*)\rangle$ given the optimal parameters — requires only polynomial classical resources. This asymmetry is structurally analogous to the NP verification paradigm that underpins all of computational complexity theory, and it provides a natural foundation for a consensus mechanism.

1.2 The Three Pillars of PoUQW

The PoUQW protocol rests on three architectural pillars:

Pillar 1: Useful Quantum Work as the Consensus Primitive. Miners — who operate or have access to quantum processing units (QPUs) — compete to solve useful parameterized quantum computational tasks drawn from an on-chain Task Registry. These tasks span molecular simulation (VQE, VQD), combinatorial optimization (QAOA), linear algebra (VQLS), quantum machine learning (quantum kernel estimation), and sampling from classically hard distributions (IQP circuits). Each completed task produces an output of genuine scientific or commercial value — a molecular ground-state energy, an optimized portfolio allocation, a logistics solution — and the proof of completion is classically verifiable by any node in the network.

Pillar 2: Quantum Randomness for Validator Selection. PoUQW replaces classical pseudo-random validator election with a quantum random beacon built on self-certifying QRNGs (SC-QRNGs). These protocols ensure that true "physics-backed" randomness is produced in a way that can be continuously certified and externally verified information theoretically with respect to its min-entropy bound – not assumed by manufacturer trust or bounded by game-theoretic rationality. A commit-reveal beacon protocol with XOR aggregation across multiple nodes ensures that a single beacon node's certified output suffices for unbiased output, avoiding the need to assume an honest participant as in RANDAO and other classic (non-quantum) commit-reveal protocols.

Pillar 3: Full EVM Compatibility. The PoUQW chain runs a standard Ethereum Virtual Machine as its smart contract execution environment, supporting Solidity, Vyper, and the entire Ethereum developer toolchain. Quantum work is used for *consensus*, not for contract execution — state transitions are deterministic and classical. Special precompiled contracts allow on-chain applications to consume quantum computation results and certified randomness, enabling novel use cases from on-chain drug discovery to provably fair gaming.

1.3 Design Philosophy and Scope

We adopt a conservative, engineering-driven design philosophy. Every mechanism in this paper is constrained to be **feasible on currently available NISQ-era quantum hardware** — devices with 50 to 1000+ noisy physical qubits, gate fidelities of 99.5%–99.8% for two-qubit operations, and circuit depths of $O(10^2)$ to $O(10^3)$ before decoherence dominates. We specifically target hardware platforms available today: IBM's Eagle (127 qubits) and Heron (133 qubits) superconducting processors, Quantinuum's H2 trapped-ion system (56 qubits with all-to-all connectivity and >99.8% two-qubit gate fidelity), and IonQ's Forte (36 algorithmic qubits). We do not invoke fault-tolerant quantum computing, logical qubits encoded in error-correcting codes, or any capability that requires hardware not yet demonstrated.

We are also intellectually honest about limitations. NISQ devices are noisy, the scale of commercially useful quantum advantage is still being established for many problem classes, and quantum hardware is expensive and controlled by a small number of vendors. Our claim is not that PoUQW delivers transformative quantum advantage today, but that it establishes the *correct architectural framework* for a blockchain that will grow in utility as quantum hardware matures — while providing genuine, if initially modest, useful computation from day one.

2 Background and Related Work

2.1 Proof-of-Work and Its Discontents

Nakamoto's Proof-of-Work [25] established the foundational paradigm for permissionless consensus: security is purchased through the expenditure of a scarce physical resource (computational work, and by extension, energy). The key insight is that the cost of attacking the network — reversing confirmed transactions via a chain reorganization — scales linearly with the honest network's cumulative computational expenditure, making attacks prohibitively expensive when the network is sufficiently large.

The limitation of Bitcoin's PoW is that the hash puzzles serve no purpose beyond consensus. Miners compute $H(nonce \| block_header) \leq target$ for a cryptographic hash function H, iterating over nonces until a satisfying preimage is found. The computational work is deliberately *unstructured* — any structure in the puzzle would potentially allow shortcuts that undermine security. This design choice means that the enormous computational resources devoted to Bitcoin mining produce no reusable scientific, commercial, or mathematical output.

Several projects have attempted to redirect PoW toward useful computation. **Primecoin** [19] replaced hash puzzles with searches for Cunningham chains and bi-twin chains of prime numbers. While mathematically interesting, the economic and scientific value of these prime chains is negligible. **Chia** [8] introduced Proofs of Space and Time, where miners prove that they have allocated disk storage to lookup tables and subsequently demonstrate elapsed time via Verifiable Delay Functions (VDFs). This reduces energy consumption but the stored data is random — it serves no useful purpose outside the consensus protocol.

More recently, proposals for **AI-based Proof-of-Useful-Work** have gained attention. These generally propose that miners perform machine learning training or inference as their consensus contribution. The most developed instantiation is Bittensor, which we analyze in detail in Section 2.4.

2.2 Proof-of-Stake and Economic Consensus

Proof-of-Stake consensus, as implemented in Ethereum 2.0 [7, 32], Cosmos/Tendermint [6], Algorand [14], and Cardano Ouroboros [18], replaces physical resource expenditure with economic collateral. Validators deposit tokens as stake and are selected to propose and validate blocks in proportion to their stake-weighted probability. Misbehavior (equivocation, invalid block proposals) is punished by *slashing* — the protocol-enforced destruction of a portion of the validator's stake.

PoS achieves strong security properties — under the assumption that a supermajority of staked capital is held by honest participants — while dramatically reducing energy consumption (Ethereum's merge to PoS reduced network energy consumption by approximately 99.95%). However, PoS has structural properties that PoUQW seeks to improve upon. First, the "work" performed by PoS validators (running the consensus protocol) is computationally trivial and produces no independently valuable output. Second, consensus power is proportional to capital, creating a wealth-concentration dynamic. Third, validator selection relies on pseudo-random mechanisms (RANDAO in Ethereum, VRFs in Algorand) whose bias resistance depends on computational assumptions, not physical laws.

2.3 Hybrid Quantum-Classical Algorithms and the Verifiability Gap

The class of algorithms central to PoUQW are **variational quantum algorithms (VQAs)** — hybrid quantum-classical methods in which a parameterized quantum circuit (the *ansatz*) prepares a trial state $|\psi(\theta)\rangle$, a classical optimizer iterates to find optimal parameters θ^*, and the quality of the solution is evaluated via a classical cost function.

Variational Quantum Eigensolver (VQE) [28] is the prototypical VQA. Given a Hamiltonian H (e.g., the electronic structure Hamiltonian of a molecule), VQE prepares trial states $|\psi(\theta)\rangle$ on a quantum circuit and measures the expectation value $\langle\psi(\theta)|H|\psi(\theta)\rangle$ through a decomposition of H into a sum of Pauli strings. The classical optimizer minimizes this expectation value over the parameter space. The key property for PoUQW is that while the *search* for θ^* requires quantum resources (the circuit prepares states in an exponentially large Hilbert space that cannot be efficiently simulated classically at sufficient scale), *verifying* the result requires only classical computation: given θ^*, one can classically compute $\langle\psi(\theta^*)|H|\psi(\theta^*)\rangle$ for small-to-moderate instances, or — for larger instances where full classical simulation is infeasible — one can verify that the submitted energy value is consistent with the measurement statistics and lies below known classical bounds.

Quantum Approximate Optimization Algorithm (QAOA) [13] applies the variational framework to combinatorial optimization. For a problem encoded as maximizing a cost function $C(\mathbf{z})$ over Ising spin assignments $\mathbf{z} \in \{-1, +1\}^n$, QAOA prepares a state through p alternating layers of problem and mixer unitaries parameterized by angles $(\gamma, \beta) \in \mathbb{R}^{2p}$. Measurement yields a bitstring z^*, and the quality of the solution is evaluated by classically computing $C(z^*)$ — a trivially efficient operation. This verification asymmetry is exact: finding high-quality z^* for hard combinatorial instances (e.g., MaxCut on random 3-regular graphs) may require quantum resources, but *checking* $C(z^*)$ is always $O(|E|)$ where $|E|$ is the number of edges in the problem graph.

This structural feature — **classically hard to produce, classically easy to verify** — is what we term the *verifiability gap*, and it is the foundation of PoUQW's consensus mechanism. The gap is analogous to the NP verification paradigm: an NP witness is hard to find but easy to check. Variational quantum algorithms, under plausible complexity-theoretic assumptions, produce witnesses (optimal parameters, high-quality bitstrings) that classical machines cannot efficiently generate but can efficiently validate.

Additional VQAs that exhibit this property and are included in the PoUQW Task Registry

are described in Section 4.1, including Variational Quantum Deflation (VQD) for excited states [17], Variational Quantum Linear Solver (VQLS) for linear systems [5], and quantum kernel methods for machine learning [16].

2.4 Quantum Random Number Generation

Ontologically indeterminate randomness — unpredictable even in principle, not merely due to incomplete knowledge — arises from quantum measurement: the outcome of measuring a quantum superposition is fundamentally undetermined until the moment of observation. The outcomes of quantum measurements on superposed states are inherently unpredictable, even in principle, by the laws of quantum mechanics. However, trusting a quantum random number generator requires some mechanism to verify that the device is actually producing quantum randomness rather than deterministic output.

Self-certifying quantum random number generators (SC-QRNGs) address this trust issue using commodity hardware. An SC-QRNG combines a physical quantum entropy source (vacuum fluctuation sampling, single-photon detection, or laser phase noise) with a statistical certification test that verifies the entropy source and computes a mathematically rigorous min-entropy bound for each output batch [4, 23]. If the measured quantum state deviates from the expected model — due to device degradation, environmental interference, or tampering — the certified entropy bound drops and the output is flagged. This provides a continuously auditable proof of randomness quality that goes well beyond trusting a manufacturer's one-time calibration. Practical randomness extraction using Trevisan's extractor or Toeplitz extractor [10, 33] can transform raw SC-QRNG outputs into uniform random bitstrings suitable for cryptographic applications.

2.5 Quantum-Resistant Versus Quantum-Native Blockchains

It is essential to distinguish PoUQW from the broader category of "quantum blockchains" discussed in the literature. Most work on quantum and blockchain intersections focuses on **quantum resistance** — hardening classical blockchain cryptography against future quantum attacks by adopting post-quantum signature schemes (e.g., CRYSTALS-Dilithium, SPHINCS+, FALCON) and post-quantum key exchange. Projects such as the QRL (Quantum Resistant Ledger) and various Ethereum post-quantum migration proposals (EIP-7702 discussions) fall into this category. These are defensive measures — they protect classical blockchains from quantum threats but do not *use* quantum computation productively.

A separate line of research investigates fully quantum consensus protocols that assume quantum communication channels and quantum memory — e.g., quantum Byzantine agreement [3] and quantum blockchain proposals based on quantum digital signatures or quantum money [9]. These assume fault-tolerant quantum hardware and quantum networking infrastructure that does not exist today.

PoUQW occupies a distinct and practical middle ground: it is a **quantum-native blockchain** that *uses* quantum computation for consensus (useful work and randomness generation) while remaining a classical distributed system at the networking and state-management layers. All

inter-node communication is classical. The quantum components are confined to (a) QPU-equipped miners who produce work proofs and (b) beacon nodes that generate certified randomness. The protocol is designed for *today's* NISQ hardware, not for hypothetical future quantum networks.

2.6 Bittensor and Proof of Useful Work for AI

Bittensor [29] is the most prominent existing implementation of a proof-of-useful-work blockchain, and its design choices, achievements, and limitations are directly relevant to PoUQW's positioning. We provide a substantive analysis below.

2.6.1 Bittensor's Design

Bittensor operates as a decentralized network of *subnets*, each dedicated to a specific AI/ML task type — text generation, image generation, text-to-speech, data scraping, protein folding prediction, and others. Within each subnet, *miners* produce AI model outputs (inferences, training contributions, or other computational results), and *validators* score the quality of these outputs using evaluation metrics and comparison functions. The native token, TAO, is distributed to miners and validators based on the assessed quality and quantity of their contributions, as determined by the Yuma Consensus mechanism.

The architecture is notable for several reasons. First, the consensus mechanism is the useful work — there is no separate hash puzzle or stake-based leader election for block production (Bittensor uses a nominated proof-of-stake substrate chain for finality, with the Yuma Consensus governing subnet reward distribution). Second, the subnet architecture allows Bittensor to encompass diverse AI task types under a single economic umbrella, with TAO emissions allocated across subnets by stake-weighted governance. Third, the system has achieved meaningful scale: as of early 2026, Bittensor operates dozens of active subnets with thousands of miners producing real AI outputs.

2.6.2 What Bittensor Gets Right

Bittensor proved several important theses that PoUQW builds upon. It demonstrated the **economic viability of replacing hash mining with productive computation** — TAO has achieved substantial market capitalization, validating the premise that a token can accrue value from the useful work its network performs rather than from artificial scarcity alone. It showed that a **decentralized marketplace for computation** can function at scale, with miners competing on output quality and validators providing a quality-assessment layer. Its subnet architecture demonstrated **extensibility** — the network can accommodate new task types without protocol-level changes, by creating new subnets with task-specific evaluation functions.

Perhaps most importantly, Bittensor established the conceptual framework that the consensus mechanism of a blockchain need not be a *cost* to the network (as in Bitcoin's energy expenditure) but can instead be the *product* of the network. This philosophical shift — from consensus-as-cost to consensus-as-product — is the foundation on which PoUQW also builds.

2.6.3 Where PoUQW Diverges

Despite these achievements, Bittensor's design contains structural features that PoUQW argues can be improved by leveraging the specific properties of quantum computation. The key divergences are:

Verification objectivity. Bittensor's most fundamental architectural challenge is that **AI output quality is inherently subjective**. When a validator scores a miner's text generation output, image, or inference result, the scoring function encodes a particular (and debatable) definition of "quality." Validators use learned metrics, comparison functions, and heuristic evaluations — not mathematical proofs of correctness. This creates several attack surfaces: miners can *game* the scoring function (optimizing for the metric rather than genuine quality — a manifestation of Goodhart's Law), validators can *collude* with preferred miners, and the quality metrics themselves may not align with end-user value. The verification of useful work in Bittensor is *opinion-based* at a fundamental level.

In PoUQW, verification is **mathematically deterministic** for the core task classes. Checking that $\langle\psi(\theta^*)|H|\psi(\theta^*)\rangle \leq E_{\text{threshold}}$ is an arithmetic operation on well-defined mathematical objects. Evaluating a QAOA cut value $C(\mathbf{z}) = \sum_{(i,j)\in E} w_{ij}(1 - z_i z_j)/2$ (with $z_i = 1 - 2b_i$ from measured bits $b_i \in \{0, 1\}$) is a summation. Computing a residual norm $\|Ax - b\|$ is linear algebra. These verifications admit no ambiguity, no subjective judgment, and no scoring function to game. The verification of quantum work in PoUQW is closer in character to Bitcoin's PoW verification ($H(x) \leq target$: either it does or it doesn't) than to Bittensor's quality scoring.

Computational asymmetry. In Bittensor, the same class of hardware (GPUs) that *produces* the AI work can also *verify* it. A validator scoring a miner's text output runs its own GPU inference to generate comparisons. There is no fundamental complexity gap between production and verification — both are polynomial-time operations on the same hardware class. This means that verification does not exploit a structural computational asymmetry, and the security of the system relies on economic incentives (validators are paid to be honest) rather than on computational hardness.

In PoUQW, the asymmetry is conjectured to be *fundamental.* Producing a high-quality VQE solution for a molecular Hamiltonian on n qubits requires preparing and measuring quantum states in a 2^n-dimensional Hilbert space — a task for which the best known classical algorithms scale exponentially for certain problem classes, while a quantum computer can efficiently prepare and measure states in this exponentially large space, with interference effects concentrating probability amplitudes toward low-energy configurations that classical methods cannot efficiently find. Verifying the result requires only $O(\text{poly}(n))$ classical operations (evaluating the Hamiltonian at the submitted parameters). This complexity-theoretic gap provides a *structural* security guarantee: even if verification were free and instantaneous, a classical adversary would still face an exponential computational barrier to forging work proofs (for sufficiently large instances under standard complexity conjectures).

Work uniqueness and physical grounding. Bittensor miners compete using commodity GPU hardware. AI inference and training are reproducible on any sufficiently powerful classical computer, and the marginal cost of GPU computation continues to decline as hardware improves. There is no *physical scarcity* inherent to the production of AI work — it is an economics of scale,

not an economics of unique physical capability.

PoUQW's useful work, by contrast, requires quantum processing units — scarce, expensive, and fundamentally different from classical hardware. A quantum computation on n qubits manipulates a state in a 2^n-dimensional Hilbert space through unitary operations that have no efficient classical equivalent for generic circuits. This means that the work *cannot be replicated* on classical machines at scale (for sufficiently complex problem instances), giving the proof-of-work a physical grounding: each $Q token represents not just useful computational output but the expenditure of a physically scarce quantum resource, analogous to the thermodynamic grounding of Bitcoin's value but channeled into productive output.

Randomness. Bittensor relies on classical pseudo-randomness for any stochastic elements of its protocol. PoUQW uses self-certifying quantum randomness from quantum hardware — a randomness guarantee backed by physics and real-time entropy certification rather than computational or game-theoretic assumptions.

Execution layer. Bittensor does not include a general-purpose smart contract execution environment. It is a purpose-built network for AI subnet coordination, lacking the composability, DeFi infrastructure, and developer ecosystem that an EVM-compatible chain provides. PoUQW includes full EVM compatibility, enabling arbitrary decentralized applications and integration with the broader Solidity ecosystem.

2.6.4 Philosophical Framing

Bittensor and PoUQW share a common philosophical lineage: both are networks where **the consensus mechanism is the productive work**. Bittensor is to artificial intelligence what PoUQW is to quantum computing — a system that converts the security budget of a blockchain into economically useful computation rather than thermodynamic waste.

The structural argument for quantum work as a consensus primitive, however, rests on the **verification asymmetry**. Scoring an AI model's output quality is inherently a judgment call — a learned function that approximates human or market preferences, subject to gaming and drift. Checking that $\langle\psi(\theta^*)|H|\psi(\theta^*)\rangle \leq E_{\text{threshold}}$ is a deterministic mathematical operation that admits no ambiguity. This makes PoUQW's security model architecturally closer to classical PoW's — where verification is objective, permissionless, and non-gameable — while retaining the useful-work property that Bittensor pioneered.

Bittensor has demonstrated something important: that a token economy grounded in useful work can function and attract meaningful value. It has also demonstrated the *challenges* of subjective verification, which manifest as ongoing disputes about scoring function fairness, subnet reward allocation, and validator collusion. PoUQW argues that quantum computation's natural verification asymmetry sidesteps these challenges by design.

The following table summarizes the key dimensions of comparison:

Dimension	Bittensor (TAO)	PoUQW ($Q)
Work type	AI inference and training	Hybrid quantum-classical computation
Verification	Subjective quality scoring by validators	Deterministic classical evaluation of mathematical cost functions
Computational asymmetry	Weak — same GPU hardware produces and verifies work	Strong — quantum production, classical verification (conjectured exponential gap)
Work uniqueness	Reproducible on commodity GPUs	Requires scarce QPU hardware; classically irreproducible at scale
Randomness source	Classical pseudo-randomness	Self-certifying quantum randomness (live entropy-certified)
Execution layer	No general-purpose smart contract VM	Full EVM compatibility
Hardware barrier	Low (commodity GPUs)	High (QPU access required for mining)
Immediate market	Large (AI/ML demand)	Small but growing (quantum computing applications)
Scoring attack surface	Goodhart's Law, validator collusion, metric gaming	Minimal — verification is arithmetic, not opinion

3 System Architecture

3.1 High-Level Overview

The PoUQW protocol is organized into five distinct but interacting layers, each responsible for a specific aspect of the system's operation. We describe each layer and its interfaces below, followed by a structural diagram.

3.2 Layer Descriptions

Layer 1: Quantum Work Layer. This is the physical substrate where quantum computation occurs. It comprises two types of quantum-equipped nodes:

QPU Miners operate hybrid quantum-classical systems. They receive task specifications from the on-chain Task Registry, execute variational quantum algorithms on their QPU hardware (with classical optimization loops running on co-located classical processors), and produce `QuantumWorkProof` artifacts that are submitted to the network. Miners may operate their own QPU hardware or access cloud-based quantum computing services (e.g., IBM Qiskit Runtime, Amazon Braket, Azure Quantum).

Beacon QRNG Nodes operate self-certifying quantum random number generators. A beacon node can be deployed by any staked validator with a quantum server and an SC-QRNG module. Beacon nodes produce entropy-certified random bits that are fed into the randomness beacon protocol.

Layer 2: Verification Layer. Classical full nodes that validate quantum work proofs. These nodes require no quantum hardware — they perform purely classical computations to verify submitted proofs: reconstructing problem instances from on-chain task specifications, evaluating objective functions at submitted solutions, performing statistical consistency checks on measurement data, and comparing results against difficulty thresholds. The verification layer is permissionless — any party with a standard classical computer can run a verification node.

Layer 3: Randomness Beacon Layer. A protocol layer that aggregates certified random outputs from multiple independent SC-QRNG beacon nodes into a single, unbiasable per-epoch

```
+-------------------------------------------------------------+
|                    APPLICATIONS & USERS                      |
|         (DApps, DeFi, Task Bounty Posters, Wallets)          |
+---------------------------+-----------------------------------+
                            | JSON-RPC / Web3
+---------------------------v-----------------------------------+
|                  EXECUTION LAYER (Layer 5)                   |
|     EVM-Compatible Smart Contract Engine (revm / evmone)      |
|     +----------------------------------------------+          |
|     | Precompiles: verifyQuantumWorkProof()          |          |
|     |              getBeaconRandomness()             |          |
|     |              getTaskResult()                   |          |
|     +----------------------------------------------+          |
|     Account-based state model | EIP-1559 gas pricing          |
+---------------------------+-----------------------------------+
                            | State transitions
+---------------------------v-----------------------------------+
|              CONSENSUS & FINALITY LAYER (Layer 4)             |
|       BFT Finality Gadget (HotStuff / Tendermint variant)     |
|    Ingests: Work Proofs + Beacon Randomness + Transactions    |
|    Outputs: Finalized blocks with deterministic state roots   |
+----------+--------------------------------+-------------------+
           |                                |
+----------v----------+          +------------v----------------+
| VERIFICATION LAYER  |          |  RANDOMNESS BEACON LAYER    |
|     (Layer 2)       |          |       (Layer 3)             |
|                     |          |                             |
| Classical nodes     |          | SC-QRNG beacon nodes        |
| validate quantum    |          | generate certified quantum  |
| work proofs:        |          | randomness:                 |
|                     |          |                             |
| * Hamiltonian eval  |          | * SC-QRNG hardware   |
| * Objective check   |          | * Min-entropy certification |
| * Statistical tests |          | * Toeplitz extraction       |
| * Threshold compare |          | * Commit-reveal aggregation |
+----------^----------+          +------------^----------------+
           | Work proofs                     | Random bits
+----------+---------------------------------+----------------+
|                QUANTUM WORK LAYER (Layer 1)                   |
|                                                               |
|  +-------------+  +-------------+  +---------------------+   |
|  |  QPU Miner  |  |  QPU Miner  |  |  Beacon QRNG Node   |   |
|  | (VQE/QAOA/  |  | (VQE/QAOA/  |  |  (SC-QRNG module,   |   |
|  |  VQD/VQLS)  |  |  VQD/VQLS)  |  |   commodity HW)     |   |
|  +-------------+  +-------------+  +---------------------+   |
|                                                               |
|          Task Registry (on-chain, governance-curated)         |
+-------------------------------------------------------------+
```

Figure 1: PoUQW five-layer architecture.

random value. The protocol uses a commit-reveal scheme where each beacon node commits to its extracted random output (with an accompanying entropy certificate) before revealing. The aggregated randomness is used to seed validator selection for each consensus epoch.

Layer 4: Consensus and Finality Layer. A BFT-style consensus protocol that combines quantum work proofs, beacon randomness, and transaction ordering into finalized blocks. Block proposers are selected using the beacon randomness, and a committee of validators (also selected by the beacon) votes to finalize blocks. The finality gadget provides deterministic, irreversible finality once a supermajority of the committee has attested to a block.

Layer 5: Execution Layer. A standard EVM-compatible virtual machine that processes smart contract transactions. State transitions are deterministic and classical — all validators execute the same transactions in the same order and arrive at the same state root. Special precompiled contracts expose quantum work results and beacon randomness to smart contract logic.

3.3 Inter-Layer Communication

All inter-layer communication uses classical networking (TCP/IP). There is no quantum communication channel between nodes. Specifically:

- Miners submit `QuantumWorkProof` structures to the P2P network as special transaction types (Layer 1 → Layer 2/4).
- Beacon nodes submit `BeaconContribution` structures containing entropy-certified random outputs (Layer 1 → Layer 3).
- Verification nodes broadcast attestations (`ProofValid` / `ProofInvalid`) (Layer 2 → Layer 4).
- The beacon protocol outputs a per-epoch `BeaconRandomness` value (Layer 3 → Layer 4).
- The consensus layer produces finalized blocks containing state transition data (Layer 4 → Layer 5).
- The execution layer processes transactions and updates the state trie (Layer 5 → state storage).

3.4 Node Types

The network supports four node types with distinct hardware requirements:

4 Proof of Useful Quantum Work — Detailed Design

The Proof of Useful Quantum Work mechanism is the core innovation of the PoUQW protocol. It defines what constitutes valid "work" for block production, how that work is represented and submitted, how it is verified by classical nodes, and how the difficulty of the work adapts over time to maintain a target block interval.

Node Type	Quantum Hardware Required	Classical Hardware	Role
QPU Miner	Yes (QPU with ≥50 qubits)	High-performance classical co-processor	Produces quantum work proofs
Beacon Node	Yes (SC-QRNG module)	Moderate	Generates certified random bits
Validator	No	Moderate-to-high (for EVM execution)	Participates in BFT consensus; executes transactions
Verification Node	No	Moderate	Validates quantum work proofs; optional full node

Table 1: Node types and their hardware requirements. Any node can fulfill multiple roles simultaneously. A QPU miner may also run a validator node if it has sufficient stake. A beacon node may also serve as a verification node.

4.1 Eligible Work Tasks: The Task Registry

The **Task Registry** is an on-chain data structure that defines the set of parameterized problem classes eligible for quantum work proofs. Each entry in the registry specifies a problem class, the quantum algorithm to be applied, the format of problem instances, the verification procedure, difficulty calibration parameters, and a usefulness weight that influences block rewards. New task classes can be added as quantum hardware matures and new variational algorithms are developed. The initial Task Registry at mainnet launch includes the following five task classes.

4.1.1 Molecular Ground-State Energy (VQE)

Algorithm. The Variational Quantum Eigensolver [28] computes the ground-state energy of a molecular Hamiltonian H. The Hamiltonian is expressed as a weighted sum of Pauli strings:

$$H = \sum_i c_i P_i, \quad P_i \in \{I, X, Y, Z\}^{\otimes n}$$

where n is the number of qubits (determined by the chosen fermion-to-qubit mapping — Jordan-Wigner, Bravyi-Kitaev, or parity). A parameterized quantum circuit (ansatz) — typically a Unitary Coupled Cluster (UCC) or hardware-efficient ansatz — prepares a trial state $|\psi(\theta)\rangle$, and the expectation value $E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle$ is estimated by measuring each Pauli string P_i and computing the weighted sum. A classical optimizer (e.g., COBYLA, L-BFGS-B, SPSA) iterates over θ to minimize $E(\theta)$.

Why it is useful. Accurate computation of molecular ground-state energies is a central problem in computational chemistry and drug discovery. It determines molecular stability, reaction energetics, binding affinities, and material properties. Classical methods (DFT, CCSD(T)) struggle with strongly correlated systems — molecules where electron-electron interactions create complex entanglement structures in the ground state. These include transition-metal catalysts, biological cofactors, and many pharmacologically relevant molecules. VQE on quantum hardware offers a path to computing these energies for systems beyond classical reach.

Why it is classically hard. Exact diagonalization of the Hamiltonian scales as $O(2^n)$ in both time and space, limiting classical methods to approximately $n \leq 50$ qubits (a few dozen

electrons in a minimal basis set). Approximate classical methods (DMRG, AFQMC) have their own limitations for strongly correlated systems. Full-configuration-interaction (FCI) quality results for systems with $n > 50$ active-space qubits are not classically feasible.

Classical verification method. Given the miner's submitted optimal parameters θ^* and the problem Hamiltonian H (reconstructed from the task specification):

1. Classically reconstruct the ansatz circuit structure.
2. For each Pauli string P_i in H, compute the expectation value $\langle\psi(\theta^*)|P_i|\psi(\theta^*)\rangle$ either by classical simulation (for $n \leq 40$) or by verifying consistency with the submitted measurement statistics.
3. Compute $E(\theta^*) = \sum_i c_i \langle\psi(\theta^*)|P_i|\psi(\theta^*)\rangle$.
4. Compare $E(\theta^*)$ against the difficulty threshold E_{thresh} and known chemical constraints (e.g., the energy must lie between the Hartree-Fock energy and the exact ground-state energy, if known).
5. Statistical checks: verify that the submitted measurement bitstrings are consistent with the claimed expectation values (chi-squared test on marginal distributions).

Verification cost. $O(N_P \cdot \text{poly}(n))$ where N_P is the number of Pauli strings in H (typically $O(n^4)$ for molecular Hamiltonians). For $n \lesssim 30$–35 qubits, full statevector simulation is feasible on high-memory servers (requiring up to $\sim$512 GB RAM); for $35 < n \leq 45$, it requires large HPC cluster resources ($\sim$16 TB RAM at $n = 40$). For $n > 40$, verification relies on statistical consistency checks and energy bound comparisons, which are $O(\text{poly}(n))$.

Difficulty calibration. The protocol adjusts difficulty by varying the problem instance complexity: the number of qubits n (number of active orbitals), the ansatz circuit depth d, and the energy threshold E_{thresh} (how close to the true ground state the solution must be). As quantum hardware improves, the protocol increases n and tightens E_{thresh} to maintain the target block time (see Section 4.4).

Anti-spoofing. A classical adversary attempting to forge a VQE proof must find parameters θ^* such that $E(\theta^*) \leq E_{\text{thresh}}$ without access to quantum hardware. For problem instances with $n > 40$ qubits in a strongly correlated regime, this requires either (a) solving the electronic structure problem classically (which is QMA-hard in the worst case — [20] — and practically intractable for strongly correlated systems), or (b) guessing parameters that happen to yield a low energy (which succeeds with negligible probability for high-dimensional parameter spaces). The difficulty threshold E_{thresh} is set below the best known classical approximation for the instance, ensuring that achieving it requires quantum resources.

Proof artifact. The miner submits: $\theta^* \in \mathbb{R}^p$ (optimized parameters), a set of M measurement bitstrings $\{z_1, \ldots, z_M\}$ from the final optimized circuit, the computed objective value $E(\theta^*)$, and QPU calibration metadata.

4.1.2 Combinatorial Optimization (QAOA)

Algorithm. The Quantum Approximate Optimization Algorithm [13] addresses combinatorial optimization problems encoded as maximization of a classical objective function $C : \{-1, +1\}^n \to \mathbb{R}$.

The canonical example is MaxCut: given a graph $G = (V, E)$ with edge weights w_{ij}, find a partition of vertices into two sets that maximizes the total weight of edges crossing the partition:

$$C(\mathbf{z}) = \sum_{(i,j)\in E} w_{ij} \frac{1 - z_i z_j}{2}, \quad z_i \in \{-1, +1\}$$

QAOA measurement yields a bitstring $\mathbf{b}^* \in \{0, 1\}^n$; the spin assignment is obtained via $z_i = 1 - 2b_i$, so $z_i = +1$ when qubit i measures $|0\rangle$ (partition A) and $z_i = -1$ when it measures $|1\rangle$ (partition B).

QAOA prepares a quantum state through p alternating layers of a problem unitary $U_C(\gamma) = e^{-i\gamma C}$ and a mixer unitary $U_M(\beta) = e^{-i\beta \sum_j X_j}$, parameterized by angles $\gamma = (\gamma_1, \ldots, \gamma_p)$ and $\beta = (\beta_1, \ldots, \beta_p)$:

$$|\psi(\gamma, \beta)\rangle = U_M(\beta_p) U_C(\gamma_p) \cdots U_M(\beta_1) U_C(\gamma_1) |+\rangle^{\otimes n}$$

Why it is useful. Combinatorial optimization problems arise across industries: portfolio optimization in finance (maximizing risk-adjusted returns subject to constraints), vehicle routing in logistics, job-shop scheduling in manufacturing, network design in telecommunications, and resource allocation in operations research. These problems are NP-hard in general, and heuristic solvers are the practical standard for large instances.

Why it is classically hard. MaxCut on random d-regular graphs is known to be NP-hard. The best classical polynomial-time approximation algorithm for MaxCut is the Goemans-Williamson (GW) semidefinite programming relaxation, which achieves an approximation ratio of $\alpha_{GW} \approx 0.878$ [15]. For QAOA at depth $p \geq 11$ on random 3-regular graphs, numerical evidence suggests performance exceeding the GW ratio [35], though rigorous proofs of quantum advantage for QAOA remain an open question.

Classical verification method. Given the miner's submitted bitstring z^* and the problem graph G:

1. Reconstruct G from the `problem_instance_hash` (fetch the full instance from an off-chain data availability layer or reconstruct from on-chain parameters).

2. Convert the submitted bitstring $\mathbf{b}^* \in \{0, 1\}^n$ to spin variables via $z_i^* = 1 - 2b_i^*$, then compute
$$C(\mathbf{z}^*) = \sum_{(i,j)\in E} w_{ij} \frac{1 - z_i^* z_j^*}{2}.$$
Equivalently, directly from bits: $C(\mathbf{b}^*) = \sum_{(i,j)\in E} w_{ij} |b_i^* - b_j^*|$.

3. Compare $C(z^*)$ against the difficulty threshold C_{thresh}.

Verification cost. $O(|E|)$ — linear in the number of edges. This is extremely efficient.

Difficulty calibration. Adjust by varying the graph size $n = |V|$, graph regularity and structure, and the threshold C_{thresh} relative to known bounds ($C_{\text{thresh}} > \alpha_{GW} \cdot C_{\max}$ to require super-GW performance, or C_{thresh} close to $C_{\max}$ for harder instances).

Anti-spoofing. For random 3-regular graph instances where C_{thresh} is set above the GW threshold, a classical adversary must find a cut exceeding the best known polynomial-time classical bound — which is believed to be computationally intractable for polynomial-time algorithms (specifically, achieving an approximation ratio beyond $\alpha_{GW} \approx 0.878$ is UGC-hard under the Unique Games Conjecture). The adversary could use classical heuristics (simulated annealing, genetic algorithms), but for carefully chosen instance families and thresholds, the expected classical runtime exceeds the block time with overwhelming probability.

Proof artifact. The miner submits: optimized QAOA angles $(\gamma^*, \beta^*) \in \mathbb{R}^{2p}$, the best bitstring z^* from M measurements, the computed cut value $C(z^*)$, and QPU calibration metadata.

4.1.3 Quantum Chemistry Excited States (VQD)

Algorithm. Variational Quantum Deflation [17] extends VQE to compute excited-state energies. After obtaining the ground state $|\psi_0(\theta_0^*)\rangle$ via VQE, VQD finds the k-th excited state by minimizing:

$$\mathcal{L}_k(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle + \sum_{j=0}^{k-1} \beta_j |\langle\psi(\theta)|\psi_j(\theta_j^*)\rangle|^2$$

where β_j are penalty weights enforcing orthogonality to previously found states; for convergence to the correct k-th excited state, each β_j must satisfy $\beta_j > E_k - E_j$.

Why it is useful. Excited-state energies determine molecular absorption spectra, photochemical reaction pathways, luminescence properties (critical for OLED design), and energy transfer processes in biological systems (photosynthesis, vision). These are essential for drug design (photodynamic therapy), materials science (solar cells, LEDs), and fundamental chemistry.

Classical verification method. (1) Verify each energy $E_k = \langle\psi_k(\theta_k^*)|H|\psi_k(\theta_k^*)\rangle$ by Hamiltonian evaluation (same as VQE). (2) Verify mutual orthogonality: $|\langle\psi_j|\psi_k\rangle|^2 \leq \epsilon$ for $j \neq k$ (computed from the submitted parameters and known ansatz structure). (3) Verify energy ordering: $E_0 \leq E_1 \leq \ldots \leq E_k$.

Verification cost. $O(k \cdot N_P \cdot \text{poly}(n))$ for k excited states, each requiring Hamiltonian evaluation plus pairwise overlap checks.

Difficulty calibration. Increase the number of excited states k required, increase molecule size n, or tighten the accuracy threshold on energies.

Anti-spoofing. Same as VQE — the miner must solve a QMA-hard problem classically, which is infeasible for sufficiently large instances.

Proof artifact. For each state k: optimized parameters θ_k^*, energy E_k, measurement bitstrings, and overlap values $|\langle\psi_j|\psi_k\rangle|^2$.

4.1.4 Linear Systems (VQLS)

Algorithm. The Variational Quantum Linear Solver [5] solves systems of linear equations $Ax = b$ variationally. The system matrix A is decomposed as a sum of unitaries $A = \sum_l c_l A_l$, and the algorithm finds parameters θ^* such that the quantum state $|\psi(\theta^*)\rangle$ encodes the *direction* of the solution — specifically, $|\psi(\theta^*)\rangle \approx |x\rangle/\|x\|$ where $x = A^{-1}b$. The state is a normalized quantum encoding of the solution vector; the actual solution x^* is recovered classically by estimating the norm $\|x\| = \|A^{-1}b\|$ separately and rescaling the amplitudes obtained from measurement. The cost function minimized is:

$$C(\theta) = 1 - \frac{|\langle b|A|\psi(\theta)\rangle|^2}{\langle\psi(\theta)|A^\dagger A|\psi(\theta)\rangle}$$

which reaches its minimum when $A|\psi(\theta)\rangle \propto |b\rangle$, achieved when

$$|\psi(\theta)\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}.$$

Why it is useful. Linearly approximated systems underlie computational fluid dynamics, structural engineering (finite element analysis), electromagnetic simulation, financial modeling, and machine learning (least-squares regression, kernel methods). Solving large sparse linearized systems is one of the most fundamental tasks in scientific computing.

Classical verification method. Given the miner's decoded solution vector x^* (extracted from measurement of $|\psi(\theta^*)\rangle$) and the problem (A, b):

1. Compute the residual $r = Ax^* - b$.
2. Compute the residual norm $\|r\|/\|b\|$.
3. Compare against the difficulty threshold: $\|r\|/\|b\| \leq \epsilon_{\text{thresh}}$.

Verification cost. $O(\text{nnz}(A))$ where $\text{nnz}(A)$ is the number of non-zero entries in A — a sparse matrix-vector multiplication.

Difficulty calibration. Adjust the system size n, condition number $\kappa(A)$ (larger κ makes the system harder to solve), and residual threshold ϵ_{thresh}.

Anti-spoofing. For systems with exponentially large condition numbers or specific structure that prevents efficient classical preconditioning, classical solvers for an N-dimensional system (encoded in $n = \log_2 N$ qubits) require $O(N^2) = O(2^{2n})$ time for direct methods, or $O(\kappa N)$ iterations for iterative methods such as conjugate gradient (where κ is the condition number), which fails to converge in polynomial time when κ grows exponentially in N. The VQLS approach encodes the solution in a quantum state, avoiding the classical memory cost of storing an n-dimensional solution vector and potentially achieving polylogarithmic scaling in system size for well-conditioned problems. For carefully chosen problem instances, the quantum approach is expected to outperform classical methods within the block time budget.

Proof artifact. Optimized parameters θ^*; the decoded solution vector x^*, reconstructed by (i) measuring $|\psi(\theta^*)\rangle$ to obtain amplitude estimates $\tilde{x} \propto x$, and (ii) rescaling by a separately estimated norm $\hat{\kappa} \approx \|A^{-1}b\|$ so that $x^* = \hat{\kappa} \cdot \tilde{x}/\|\tilde{x}\|$; the residual norm $\|Ax^* - b\|/\|b\|$; and QPU calibration metadata.

4.1.5 Quantum Machine Learning Feature Maps (Quantum Kernel Estimation)

Algorithm. Quantum kernel methods [16, 31] use parameterized quantum circuits to embed classical data into high-dimensional Hilbert space and estimate kernel values $k(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2$, where $|\phi(x)\rangle = U(x)|0\rangle^{\otimes n}$ is the quantum feature map. These kernels can then be used in classical support vector machines (SVMs) or other kernel methods for classification, regression, and anomaly detection.

Why it is useful. Quantum kernels may provide computational advantages for specific data distributions where the quantum feature map induces a kernel function that is classically intractable to evaluate [22]. Applications include molecular property prediction (using quantum-native feature maps on molecular descriptors), financial fraud detection, and materials property classification.

Classical verification method. The verifier selects a random subset $S \subset \{(i, j)\}$ of data point pairs from the submitted kernel matrix and independently computes $k(x_i, x_j)$ using classical simulation of the quantum feature map circuit (feasible for $n \leq 40$ qubits) or by re-executing on an independent QPU (for larger instances). Statistical agreement between the subset and submitted values is assessed via a mean absolute error bound $\max_{(i,j)\in S} |k_{\text{submitted}}(x_i, x_j) - k_{\text{verified}}(x_i, x_j)| \leq \delta$ for a protocol-specified tolerance δ, certifying consistency.

Verification cost. $O(|S| \cdot 2^n)$ for full simulation verification, or $O(|S|)$ for QPU-based spot checks. For NISQ-scale circuits ($n \leq 40$), classical simulation of the subset is feasible within seconds.

Difficulty calibration. Adjust the number of data points N, feature map circuit depth, qubit count n, and required kernel matrix precision.

Anti-spoofing. For quantum feature maps with $n > 40$ qubits and sufficient circuit depth, classical simulation of the full kernel matrix is exponentially costly. However, we note that the quantum advantage argument for kernel methods is more nuanced than for VQE/QAOA, and depends on the specific data distribution.

Proof artifact. The feature map circuit specification, data point set $\{x_i\}$, computed kernel matrix K_{ij}, and QPU calibration metadata.

4.2 Work Proof Format

A miner's `QuantumWorkProof` submission contains four groups of fields: *task identification* (task class ID, Keccak-256 hash of the problem instance, URI to the full instance data); *computation specification* (algorithm enum, circuit depth, qubit count, ansatz template hash); *solution data* (optimized parameters θ^*, measurement bitstrings, shot count M, claimed objective value, and — for optimization/linear-solver tasks — the explicit solution bitstring or decoded vector); and *provenance* (QPU device ID, calibration snapshot with gate fidelities and T_1/T_2 times, execution timestamp, miner address, CRYSTALS-Dilithium signature over the entire proof, and block/epoch metadata). The provenance fields feed statistical anti-spoofing heuristics and dispute resolution but cannot be cryptographically verified without trust in the QPU provider.

4.3 Difficulty Adjustment

PoUQW targets a 12-second block interval using a dual-timescale adaptive mechanism. A *fast adjustment* runs per block: it computes the average block time over a trailing 256-block window, forms the ratio $r = T_{\text{observed}}/T_{\text{target}}$, and applies a dampened correction ($\alpha = 0.1$) to objective thresholds — relaxing them when blocks are slow, tightening when fast. A *slow adjustment* runs per epoch (every 2,048 blocks, 6.8 hours): if blocks are consistently too fast, the protocol raises the structural floor (minimum circuit depth, qubit count) and tightens thresholds further ($\alpha = 0.25$); if too slow, it relaxes them down to a hard minimum. At each epoch boundary, active problem instances are rotated using beacon randomness, preventing pre-computation. The combined effect is a self-regulating policy: difficulty tracks the quantum hardware frontier, ensuring each $Q token always represents state-of-the-art computation.

The key design principles of this difficulty adjustment are:

1. **Dual-timescale stability.** The fast adjustment handles short-term fluctuations (a large QPU temporarily going offline, burst of mining activity), while the slow adjustment handles structural shifts (new quantum hardware becoming available, hardware decommissioning).

2. **Problem instance rotation.** At each epoch boundary, the set of active problem instances is rotated using the quantum beacon randomness as a seed. This prevents miners from pre-computing solutions to known instances during off-peak times and submitting them during the next epoch. Fresh instances ensure that work is performed on-demand.

3. **Hardware-tracking thresholds.** As quantum hardware improves (higher qubit counts, better gate fidelities, longer coherence times), the protocol automatically increases difficulty — more qubits, deeper circuits, tighter thresholds — ensuring that each block's work proof always represents a computation near the frontier of current quantum capability. The token never becomes "easy to mine."

4. **Graceful degradation.** If quantum hardware becomes temporarily scarce (e.g., a major cloud QPU provider experiences an outage), the difficulty relaxes automatically, maintaining chain liveness. The minimum difficulty floor ensures that proofs always require genuine quantum computation above a baseline complexity.

5 Quantum Random Beacon for Validator Selection

5.1 Self-Certifying Quantum Randomness

The PoUQW randomness beacon exploits a property unique to quantum physics: measurement outcomes on quantum systems are *ontologically* random — not merely unpredictable due to classical ignorance, but fundamentally indeterminate until the moment of measurement. Unlike classical PRNGs (which are deterministic algorithms seeded with entropy) or classical physical noise sources (which are chaotic but in principle deterministic at a microscopic level), quantum entropy sources derive randomness from processes whose indeterminacy is guaranteed by the structure of quantum mechanics itself.

5.1.1 Self-Certifying QRNGs

The source of entropy in a quantum system may take different forms, such as: vacuum fluctuation sampling (measuring the shot noise of the electromagnetic vacuum via homodyne detection), single-photon path detection (sending single photons through a beamsplitter and recording which detector fires), or laser phase noise (interfering a laser pulse with a delayed copy and measuring the inherently random quantum phase diffusion). All these three mechanisms produce randomness rooted in quantum indeterminacy.

This quantum indeterminacy suggests the possibility of random number generators whose unpredictability is guaranteed not by computational assumptions but by the laws of physics themselves. Fully manifesting the promise of quantum randomness requires confronting a fundamental challenge: how can we certify that a device's claimed quantum randomness is genuine, particularly when the device itself might be faulty or miscalibrated? A **self-certifying quantum random number generator (SC-QRNG)** combines the physical quantum entropy source with a real-time monitoring system that verifies the quantum nature of the randomness being produced. There exist various certification protocols, each offering different trade-offs between security strength and practical feasibility.

The "golden standard" of such certification methods is based on Bell's theorem, which proves that any local hidden variable theory must satisfy certain inequalities that quantum mechanics can violate. Existing implementations of loop-free tests use the CHSH inequality, which constrains the correlations that can exist between two distant parties measuring entangled quantum particles in the classical (non-quantum) setting. These measurements operate like a public certificate that can be statistically tested to ensure the proposed output bits are random in the deepest sense that physics allows, resulting in what is referred to as *device-independent (DI)* certified randomness. Yet, this method comes at a steep practical cost (laboratories separated by kilometers, high-efficiency single-photon detectors operating at cryogenic temperatures, etc.)

A more practical alternative is based on the so-called *Semi-Device-Independent (SDI)* protocols. This approach makes limited, physically reasonable assumptions about the quantum system while still leaving the most vulnerable components untrusted. The result is a class of randomness generators that achieve information-theoretic security without requiring the extreme experimental conditions of fully device-independent protocols.

Several distinct physical approaches fall within the semi-device-independent framework, each suited to different hardware capabilities and security requirements. Source-independent protocols, perhaps the most mature of these approaches, assume a trusted measurement device while treating the quantum state preparation as completely untrusted. The security of these protocols derives from the fundamental quantum noise properties of coherent detection. Even if an adversary controls the source, sending carefully crafted quantum states designed to minimize unpredictability, the Heisenberg uncertainty principle ensures that certain measurements will always retain a minimum level of irreducible noise. This noise floor, typically quantified as half a quantum of action, provides a lower bound on the min-entropy of the measurement outcomes that can be certified without any knowledge of the input state.

Measurement-device-independent protocols invert this trust structure, assuming a trusted state preparation while treating the measurement apparatus as potentially malicious. These

protocols typically employ entangled photon pairs and coincidence detection, enabling the certification of randomness even when the detectors themselves are untrusted or controlled by an adversary. The security derives from the correlations between measurement outcomes, which must satisfy certain bounds if the underlying physics is quantum mechanical.

A particularly elegant approach for single quantum systems uses temporal rather than spatial correlations. The Leggett-Garg inequality (LGI) extends Bell's framework to measurements performed on the same quantum system at different times, testing whether the system's evolution is consistent with macrorealism, the classical notion that a system possesses definite properties at all times, whether measured or not. Violations of the Leggett-Garg inequality, which can be observed in a single qubit evolving under coherent control, prove that measurement outcomes at later times cannot be predicted from knowledge of earlier measurements. This temporal certification eliminates the need for spatial separation entirely, enabling protocols that run on a single quantum device.

5.2 Certification and Externally Verifiable Security

Semi-device-independent protocols enable genuine external verification. The certification is transparent, reproducible, and can be performed by anyone with access to the raw measurement data and circuit parameters. This verifiability stands in stark contrast to commercial devices, which provide only internal health indicators without externally auditable proofs of quantum origin.

The verification process begins with the publication of raw measurement outcomes, timestamps, and experimental parameters. Using standard statistical methods, any party can compute the relevant certification metric, whether it be the Leggett-Garg inequality violation, the dimension witness value, or the heterodyne detection variance. These statistics provide a direct, mathematically rigorous proof of quantum behavior that does not require trusting the device manufacturer. At the end of this certification process, any verifier can be assured about an absolute minimum entropy bound $H_{\min}^{\text{live}}$.

For blockchain and distributed applications, this external certification can be integrated directly into smart contracts. The contract can verify that the observed statistics exceed the classical bound before accepting the randomness into the system. This on-chain verification creates an immutable, publicly auditable record of the quantum certification, ensuring that every block's randomness was genuinely certified at the time of generation.

The security provided by these protocols is information-theoretic, meaning it holds against adversaries with unlimited computational power, including quantum computers. This contrasts fundamentally with computational security, which relies on the assumed hardness of mathematical problems that future algorithms might solve. The randomness certified by a Bell or Leggett-Garg violation is unpredictable, not because we do not know how to predict it, but because the laws of quantum mechanics forbid any prediction, even in principle.

5.2.1 Randomness Extraction

Raw QRNG outputs contain genuine quantum randomness but may not be uniformly distributed — the raw bit stream includes predictable components arising from the measurement appara-

tus (detector bias, electronic noise floor, digitization artifacts). To produce cryptographically uniform random bits, the raw output is processed through a **quantum-proof randomness extractor**.

PoUQW uses **Toeplitz's extractor** [21, 30], which has been proven secure against quantum side-information. Given n raw bits with certified min-entropy k (as reported by the SC-QRNG's live monitoring), a Toeplitz extractor produces $m \leq k - 2\log(1/\epsilon)$ bits that are ϵ-close to uniform, even against an adversary holding quantum side-information. The extractor requires a short seed that can be sourced from the previous epoch's beacon output in practice (bootstrapped from a hardcoded genesis seed for the first epoch). We prefer a Toeplitz extractor over other alternatives (e.g., Trevisan [10, 33]) due to its much better practical performance.

The extraction process for each beacon node:

1. Accumulate N_{raw} raw bits from the SC-QRNG during the epoch.
2. Collect the device's live min-entropy certificates $\{H^{\text{live}}_{\min,j}\}$ for each output batch j.
3. Compute the aggregate certified min-entropy: $H^{\text{total}}_{\min} = \sum_j H^{\text{live}}_{\min,j}$.
4. Apply Toeplitz extractor with the epoch seed, extracting $m = 256$ near-uniform bits.
5. Output: 256 extracted random bits plus the entropy certificates as the beacon contribution.

The extraction process can be illustrated with a concrete example. Consider a Leggett-Garg protocol running on a cloud quantum computer, where the observed violation of the temporal inequality certifies a certain amount of min-entropy in the raw measurement outcomes. For a sequence of one hundred thousand raw measurements with a certified Leggett-Garg parameter of 1.3, the quantum conditional min-entropy might be bounded at approximately sixty thousand bits. Applying a Toeplitz extractor with a security parameter of 2^{-128} yields nearly 59K uniform bits, achieving an extraction efficiency exceeding ninety-nine percent.

5.2.2 Security Assumptions

The SC-QRNG beacon model rests on the following explicit security assumptions, ordered from strongest to weakest:

Assumption 1: Quantum mechanics is correct. The entropy source produces genuine quantum randomness as described by the quantum mechanical model of the device (e.g., vacuum fluctuations obey shot-noise statistics, beamsplitter outputs follow Born's rule). This is the same foundational assumption underlying all quantum technologies and has been verified to extraordinary precision across a century of experiments. We consider this assumption uncontroversial.

Assumption 2: The measurement model is faithful. Semi-device-independent approaches make assumptions about the measurement model on the device, such as dimension bound or source-independence. The dimension-bound assumption assumes that the quantum system being measured lives in a Hilbert space of bounded dimension, typically two for a single qubit. If the actual dimension exceeds this bound, an adversary could hide information in higher-order modes that the protocol does not monitor. We consider this a natural assumption

for systems designed to operate as fixed qubits. Source-independent protocols require trusting that the heterodyne or homodyne detector must perform as specified, with the quantum-limited noise floor that provides the security guarantee. While the source itself is untrusted, the measurement apparatus must be calibrated and free from trojans that could leak information or manipulate outcomes. This trust in the measurement device is a relaxation from device independence, but it is far weaker than the complete trust required by "conventional" quantum random number generators.

Periodic verification through **quantum state tomography** can provide additional assurance that the underlying assumptions remain valid. The device periodically diverts a fraction of its quantum signal into a monitoring channel that performs tomographic reconstruction of the quantum state being measured. By comparing the reconstructed state against the expected quantum model (e.g., verifying that vacuum fluctuation statistics match the predicted shot-noise distribution, or that photon detection statistics match the expected binomial distribution for a balanced beamsplitter), the device computes a **real-time min-entropy bound** — a mathematically rigorous lower bound on the entropy per output bit:

$$H_{\min}^{\text{live}}(X|E) \geq g(\hat{\rho}_{\text{tomo}}, \delta)$$

where $\hat{\rho}_{\text{tomo}}$ is the tomographically reconstructed state, E is classical side-information, and δ is a confidence parameter. If the entropy bound drops below a threshold (indicating device degradation, environmental interference, or tampering), the device raises an alarm and its output is rejected by the beacon protocol.

While full tomography at each beacon generation is too slow for production use, requiring many measurements at different bases to reconstruct the quantum state, periodic discrete tomography serves as a valuable calibration tool. Running full state tomography daily or weekly can verify that the quantum system is operating within expected parameters, detect drift in the machine's calibration, and provide early warning of any degradation in performance.

Assumption 3: The device firmware has not been compromised. The SC-QRNG's monitoring logic (tomographic reconstruction, entropy estimation, alarm thresholds) executes correctly. A hardware Trojan or compromised firmware could in principle spoof the monitoring output while substituting deterministic data for the quantum entropy. This is the most significant practical assumption and the primary difference from a fully device-independent (Bell-test-based) scheme, where randomness is certified without *any* device trust.

Mitigations for Assumption 2 and Assumption 3:

- **XOR aggregation.** The beacon output $R_e = \bigoplus_i r_i$ is unbiased as long as *at least one* beacon node's QRNG is functioning honestly. An adversary must compromise *every* beacon node's hardware to bias the output.

- **Multi-vendor diversity.** The protocol requires beacon nodes to use QRNGs from at least two distinct manufacturers. A firmware compromise would need to simultaneously affect devices from independent supply chains.

- **Third-party audit and open firmware.** Beacon node operators are incentivized to use QRNGs with open-source firmware and published hardware schematics, enabling in-

dependent audit. The protocol's governance can maintain an approved device list based on audit status.

- **Statistical cross-validation.** The beacon protocol performs inter-node statistical consistency checks: if all B beacon nodes are producing genuine quantum randomness, their outputs should be statistically independent. Detectable correlation between beacon contributions triggers an alert.

5.3 Deployment Options

The deployment of SC-QRNG offers multiple pathways, including cloud-based services or dedicated hardware. As a starting point, there exist commercial implementations based on fully trusted quantum hardware modules, such as Quantum Dice's DISC™ technology, which uses integrated quantum tomography to certify entropy in real time on a chip-scale device, and ID Quantique's Quantis family, which provides NIST SP 800-90B-certified entropy sources with hardware health monitoring. These devices are available as USB, PCIe, or chip-integrated modules at estimated price points of $1K–$10K — commodity hardware deployable by any staked validator. Moving to systems with more relaxed trust assumptions, a custom heterodyne-based source-independent system using commodity telecom components—1550 nm lasers, optical hybrids, balanced photodiodes, and FPGA processing—can be assembled for an estimated $5K–$17K, potentially achieving megabit-per-second rates with SDI security.

On the cloud-based front, recent advances [22, 26] have demonstrated that such protocols can run on commercially available cloud quantum computers, eliminating the need for purchasing and maintaining specialized quantum hardware. This development brings certifiable quantum randomness within reach of any organization with internet access and modest computational resources for verification. Alternatively, there exist publicly available service providers that can be used, such as NIST's CURBy beacon, which provides freely accessible 512-bit pulses every minute based on loophole-free Bell tests (the "golden standard" of certified quantum randomness) and commercial APIs from Outshift and LfD that offer trusted-device quantum randomness at no cost for evaluation tiers.

A pragmatic architecture for a blockchain-integrated system uses multiple independent backends for redundancy and distributed trust. All the heterogeneous sources—cloud LGI for distributed verification, NIST CURBy for device-independent anchor randomness, and on-premises heterodyne or commercial hardware for high-throughput independence—create a resilient system where the compromise of any single source cannot undermine the collective randomness.

5.4 Beacon Protocol

The beacon protocol aggregates randomness from multiple independent SC-QRNG beacon nodes into a single per-epoch random value used for validator selection. The protocol is designed to be **unbiasable** (no single participant can influence the output), **unpredictable** (the output cannot be known before reveal), and **verifiable** (any node can check that the beacon output was correctly derived from certified QRNG data).

5.4.1 Epoch Structure

Time is divided into **epochs** of $N_{\text{epoch}} = 2048$ blocks (approximately 6.8 hours at a 12-second block time). Each epoch has a fixed validator committee selected by the previous epoch's beacon output. The beacon protocol for epoch e proceeds in three phases:

Phase 1: QRNG accumulation (first 80% of epoch $e-1$). Each registered beacon node continuously generates random bits from its SC-QRNG and accumulates entropy certificates. The number of registered beacon nodes is B (minimum $B_{\min} = 3$ for liveness; target $B = 7$–15 for robust decentralization).

Phase 2: Commit (last 15% of epoch $e-1$). Each beacon node i runs the Toeplitz extractor on its accumulated raw output to produce $r_i \in \{0,1\}^{256}$ and assembles its entropy proof $\pi_i = (H^{\text{total}}_{\min,i}, \text{device_attestation}_i, \text{tomo_digest}_i)$. It publishes a commitment $C_i = \text{Hash}(r_i \| \pi_i \| \text{salt}_i)$ as a transaction on-chain.

Phase 3: Reveal (last 5% of epoch $e-1$). Each beacon node reveals $(r_i, \pi_i, \text{salt}_i)$ on-chain. Validators verify: (a) the commitment opens correctly; (b) the entropy proof shows $H^{\text{total}}_{\min,i} \geq H_{\min,\text{required}}$ (a protocol parameter, e.g., ≥ 512 bits of certified min-entropy for 256 bits of extracted output); (c) the device attestation is from an approved QRNG on the governance-maintained device list.

Aggregation. The epoch e beacon output is computed as:

$$R_e = \bigoplus_{i \in \mathcal{B}_{\text{valid}}} r_i$$

where $\mathcal{B}_{\text{valid}}$ is the set of beacon nodes that submitted valid commitments and reveals, and $\oplus$ denotes bitwise XOR. The XOR aggregation ensures that the output is unbiased as long as *at least one* beacon node produced genuine random bits — a single honest beacon node suffices for randomness quality.

5.4.2 Comparison to existing approaches for randomness generation

Existing approaches for generating randomness for blockchain protocols typically fall under two categories: commit-reveal protocols, such as Ethereum's RANDAO, and those based on specific cryptographic primitives like verifiable random functions (e.g., used by Chainlink, Algorand), or threshold signatures (e.g., used by Drand). Regarding the latter, existing deployed techniques are not post-quantum secure, and protocols that are secure against quantum adversaries are less studied and far from practical.

Ethereum's RANDAO mechanism (used in its PoS Beacon Chain) aggregates randomness by XOR-ing random values committed and subsequently revealed by block proposers. Clearly, there is no way to guarantee the true random nature of the final value, as it is impossible to verify how/whether individual proposers sampled their claimed random values. Instead, RANDAO and all similar protocols need to assume the existence of at least one honest proposer who is furthermore able to then reveal its committed randomness.

PoUQW's SC-QRNG beacon provides a stronger guarantee rooted in physics. Each beacon node's randomness derives from a physical quantum process whose entropy is continuously certified by real-time monitoring. Unlike RANDAO, where a validator can freely choose any value

as their "random" contribution, a beacon node with a functioning SC-QRNG *cannot* control its output without physically tampering with the measuring module of the device or during post-processing in a way that would be detected by the monitoring system or by inter-node statistical cross-validation. Furthermore, the final XOR aggregation ensures that even if some beacon nodes are compromised, a single well-functioning node suffices for unbiased output.

One issue that is common between RANDAO and our approach is the so-called **last-revealer attack**: the last proposer in the reveal phase can observe all other reveals and decide whether to reveal their own value or forfeit their proposal reward. Importantly, even if the last revealer chooses not to reveal, the beacon output is still unbiased (just computed over the remaining $B - 1$ contributors). However, the last revealer effectively gains one bit of influence over the randomness output per epoch. Our mitigation for this is adopted by RANDAO: proposers are economically motivated to open their commitments, as the cost to a non-revealing node is forfeiture of beacon rewards and slashing of their beacon node stake.

5.5 Validator Election

5.5.1 Staked Validator Set

Validator participation requires staking $Q tokens. Validators deposit tokens into a staking contract, enter an activation queue, and become eligible for committee selection after a minimum bonding period. Stake determines voting weight in BFT consensus and selection probability.

The validator set is partitioned into two roles: **block proposers** (selected per-slot to assemble and propose blocks) and **attestors** (selected per-slot to validate and vote on proposed blocks).

5.5.2 Deterministic Selection from Beacon Output

At the beginning of each epoch e, the beacon output R_e deterministically fixes the full validator schedule for all slots in the epoch. For each slot t, the proposer and attestor committee are computed as:

$$\text{Proposer}(t) = \text{StakeWeightedSelect}(\text{Hash}(R_e \| t \| \texttt{"proposer"}), \text{stake_table})$$

$$\text{Committee}(t) = \text{StakeWeightedSample}(\text{Hash}(R_e \| t \| \texttt{"committee"}), \text{stake_table}, C)$$

where $C = 128$ is the committee size and selection probability is proportional to each validator's stake s_v / S_{total}. The schedule is fully deterministic — given R_e and the stake table, any node can independently compute the proposer and committee for every slot.

Because the beacon output derives from certified quantum randomness aggregated via XOR, no party can influence who is selected. The entire schedule for the epoch is fixed the moment R_e is finalized; there is no grinding attack and no adaptive manipulation.

6 Consensus Protocol

6.1 Block Production

A block is valid if and only if it contains: (1) a `QuantumWorkProof` passing the verification protocol under the current difficulty parameters; (2) a valid set of transactions; (3) the proposer's address matching the beacon-derived schedule for this slot; and (4) consistent beacon epoch and randomness references. Standard validity conditions (correct state root, temporal ordering, signature checks) follow Ethereum conventions.

6.2 BFT Finality

PoUQW uses a **HotStuff**-derived BFT finality gadget [36] with standard three-phase commit (prepare → pre-commit → commit). The committee for each slot is selected by the quantum beacon (Section 5.3). Finality is achieved when $\geq 2f + 1$ committee members (where $f = \lfloor (C-1)/3 \rfloor$, $C = 128$) have voted through all three phases — typically within a single slot ($\approx$ 3–6 seconds). Safety and liveness follow from standard HotStuff proofs; the only PoUQW-specific additions are that attestors must verify the quantum work proof as part of block validation, and that proposer timeout rotates to the next beacon-selected candidate.

6.3 Fork Choice Rule

Before finalization, transient forks are resolved by selecting the chain with the greatest **cumulative useful work**:

$$W(\text{chain}) = \sum_{b \in \text{chain}} \text{usefulness_score}(b.\text{work_proof})$$

where usefulness_score weights the quantum work proof by task class difficulty and solution quality. This is the quantum analog of Bitcoin's longest-chain rule, but measured in useful computational output rather than hash iterations.

6.4 Incentives and Tokenomics Overview

This section provides a structural overview of the incentive mechanism; the full economic model and value thesis are developed in Section 8.

6.4.1 Block Rewards

The block reward for block b is:

$$R_b = R_{\text{base}} \cdot \mu(\text{task_class}_b) \cdot q(v_b, v_{\text{thresh},b})$$

where:

- R_{base} is the base block reward (following a halving schedule analogous to Bitcoin's, with an initial emission rate calibrated to the target total supply).

- μ(task_class) is the **usefulness multiplier** for the task class, set by governance. More commercially or scientifically valuable tasks (e.g., VQE for drug-relevant molecules) receive higher multipliers than benchmark tasks (e.g., random circuit sampling).
- $q(v, v_{\text{thresh}})$ is the **quality factor**: $q = 1+\alpha\cdot(v-v_{\text{thresh}})/v_{\text{thresh}}$ for solutions exceeding the threshold (capped at $q_{\max}$), incentivizing miners to find better solutions, not just solutions that barely pass.

The block reward is distributed as follows: 70% to the QPU miner who produced the work proof and 30% to the block proposer (who may or may not be the miner).

6.4.2 Staking Rewards and Slashing

Attestors earn staking rewards proportional to their uptime and correct voting participation. The slashing model follows Ethereum's Casper FFG design with an additional quantum-work-specific condition. Slashable offenses are equivocation (double-voting) and surround voting, each carrying an initial penalty of 1/32 of the offender's effective balance. On top of this minimum, a **correlated penalty** scales with the total stake slashed in the surrounding ±18-day window: if the fraction of total staked $Q slashed in that window is p, the correlated penalty is min($3p$, 1) of the offender's effective balance, ensuring the total slash (initial penalty plus correlated penalty) does not exceed the validator's full stake. This ensures that isolated accidents are lightly penalized while coordinated attacks destroy the attackers' capital — identical to Ethereum's approach. Slashed validators enter a forced exit queue and cannot withdraw for a 36-day period, during which the correlated penalty is assessed. Additionally, proposing a block with a quantum work proof that fails verification incurs a 1/32 penalty and immediate ejection from the active set, analogous to Bittensor's deregistration of underperforming miners from subnet slots.

6.4.3 Task Bounty Market

External parties (pharmaceutical companies, logistics firms, financial institutions, researchers) can post **task bounties** on-chain: specific problem instances accompanied by a $Q-denominated reward for the first miner to solve them at a specified quality threshold.

```
TaskBounty {
    poster_address:     address
    task_id:            bytes32     // Task class from the registry
    instance_data:      bytes       // Full problem specification
    quality_threshold:  int256      // Required objective value
    reward:             uint256     // $Q reward for solution
    expiry_block:       uint64      // Bounty expiration
    access_control:     bytes       // Optional: encryption key or access NFT for result
}
```

When a miner produces a work proof that solves a bounty, the bounty reward is transferred to the miner in addition to the standard block reward. If the bounty specifies access control, the

solution is encrypted and accessible only to the poster (or holders of the access NFT), creating a marketplace for private quantum computation results.

6.4.4 EVM Execution Layer

PoUQW runs a standard EVM (based on `revm`) with full Cancun-spec compatibility. Solidity, Hardhat, Foundry and the entire Ethereum toolchain work without modification. Existing mainnet contracts deploy as-is. The state model, account structure, and transaction execution semantics are identical to Ethereum's, with one change: EOA signatures use CRYSTALS-Dilithium (post-quantum) rather than ECDSA. Quantum work is for consensus only — smart contract execution is entirely classical and deterministic.

Gas pricing follows EIP-1559 (base fee burn + priority tip), entirely separate from quantum work rewards. The block gas limit is initially 30M gas, adjustable by governance.

A natural future direction is **quantum-aware precompiled contracts** that expose beacon randomness and quantum work results directly to smart contract logic. A `getBeaconRandomness(epoch)` precompile, for instance, would give any on-chain application access to physics-certified unbiasable randomness — enabling provably fair lotteries, gaming, NFT minting, and dispute resolution with guarantees strictly stronger than any classical oracle. Similarly, precompiles for verifying quantum work proofs and querying task results on-chain would allow DeFi protocols, DAOs, and bounty contracts to natively consume quantum computation outputs.

7 Use Cases

This section describes eight concrete use cases that illustrate how the combination of useful quantum work, certified randomness, and EVM-compatible smart contracts enables applications that are not possible — or are strictly inferior — on existing blockchain platforms.

7.1 Drug Discovery as Mining

The pharmaceutical industry faces a fundamental computational bottleneck: accurately predicting the electronic structure of drug candidate molecules and their interactions with biological targets requires solving the quantum many-body problem — a task that is exponentially hard for classical computers for strongly correlated systems. Current computational chemistry methods (DFT, coupled cluster) involve uncontrolled approximations that can miss critical phenomena such as transition-state energies, non-covalent interactions, and spin-state splittings in metalloenzyme active sites.

In PoUQW, pharmaceutical companies and research institutions can post **molecular Hamiltonian bounties**: specific molecules or molecular fragments for which ground-state and excited-state energies are needed. The problem Hamiltonian is published on-chain (encoded in the Task Registry format), accompanied by a $Q-denominated reward for solutions meeting a specified accuracy threshold.

QPU miners compete to solve these Hamiltonians using VQE and VQD, producing ground-state energies, excited-state spectra, and optimized molecular geometries. The results are verified on-chain and, depending on the bounty's access control settings, either published as a public

good (contributing to an open-access quantum chemistry database) or encrypted for the bounty poster's exclusive use.

A concrete scenario: a biotech startup studying a novel kinase inhibitor posts the active-site Hamiltonian (a 40-qubit model of the binding pocket with the candidate drug molecule) as a task bounty with a 500 $Q reward. Over the course of several blocks, miners compete to find the lowest VQE energy. The winning miner's result — the binding energy of the drug candidate — is published on-chain, and the bounty is paid. The startup uses this data to prioritize wet-lab synthesis candidates, accelerating their drug development pipeline by weeks compared to waiting for scarce QPU time from a cloud provider.

7.2 Financial Portfolio Optimization

Portfolio optimization — the problem of allocating capital across assets to maximize risk-adjusted returns subject to constraints — is a quadratic optimization problem that becomes combinatorial when discrete constraints are introduced (e.g., integer lot sizes, cardinality constraints, sector limits). Classical solvers handle moderate instances well but struggle with large portfolios (> 100 assets) with complex constraint sets.

DeFi protocols on PoUQW can post portfolio optimization problems as QAOA task bounties. A decentralized fund management protocol specifies a universe of assets (represented as an on-chain token list), a covariance matrix (updated via oracle feeds), and a set of constraints. Miners solve the combinatorial optimization via QAOA, producing an allocation bitstring z^* that maximizes the risk-adjusted return objective. The verified result is consumed on-chain by the fund management contract via the `getTaskResult` precompile, which automatically rebalances the fund's positions.

A concrete scenario: a DAO-governed yield aggregator posts a weekly rebalancing optimization over 64 DeFi yield sources with risk, liquidity, and concentration constraints. The QAOA solution, verified on-chain, is executed atomically by the rebalancing contract — no off-chain oracle, no centralized solver, no trust assumption beyond the quantum work verification.

7.3 Provably Fair Randomness for Gaming and Lotteries

Online gaming and lottery platforms face a persistent trust deficit: users must trust that the operator or random number oracle has not manipulated outcomes. Existing solutions (Chainlink VRF, commit-reveal schemes) provide computational fairness guarantees — the randomness is unbiased assuming the underlying cryptographic assumptions hold. But cryptographic assumptions can be wrong, and oracle operators can collude.

PoUQW's SC-QRNG beacon provides a stronger guarantee: the randomness consumed by gaming and lottery smart contracts derives from a **physical quantum process with continuous entropy certification**. Each beacon node's output is accompanied by a real-time certification of its produced randomness, and the XOR aggregation across multiple nodes means a single honest beacon suffices for unbiased output. Unlike Chainlink VRF (where trust is placed in a threshold of oracle nodes running deterministic algorithms), PoUQW's randomness is rooted in quantum indeterminacy — a property of nature, not a computational construction.

A concrete scenario: a decentralized lottery contract on PoUQW uses `getBeaconRandomness(currentEpoch)` to draw winning numbers. The contract verifies that the beacon's aggregate entropy certification meets the protocol threshold before proceeding. Users can independently verify — by checking the on-chain entropy certificates and device attestations — that the randomness was generated by certified quantum hardware. No operator, oracle, or validator coalition can bias the outcome without simultaneously compromising every beacon node's SC-QRNG hardware.

7.4 Supply Chain and Logistics Optimization

Global supply chains involve combinatorial optimization at massive scale: vehicle routing (minimizing total distance or time for a fleet of vehicles visiting a set of locations), warehouse allocation, production scheduling, and multimodal freight optimization. These are NP-hard problems for which classical heuristics (simulated annealing, genetic algorithms, branch-and-bound) provide good but often suboptimal solutions.

Logistics companies can post routing and scheduling problems as QAOA task bounties on PoUQW. The problem graph (locations, distances, constraints) is encoded as a combinatorial optimization instance, and miners compete to find high-quality solutions. Verified solutions are published on-chain and can be consumed by supply chain management smart contracts that coordinate between multiple parties (shippers, carriers, warehouses) in a trustless manner.

A concrete scenario: a decentralized freight marketplace posts a daily vehicle routing problem for 50 delivery locations across a metropolitan area. The QAOA solution — an optimized route sequence — is verified on-chain and automatically dispatched to carrier wallets via the logistics smart contract. The optimization is performed by the PoUQW mining network, eliminating the need for the marketplace to operate its own optimization infrastructure or trust a third-party solver.

7.5 Quantum Machine Learning

Quantum kernel methods offer a potential advantage for specific classification and regression tasks where the quantum feature map induces a kernel function that captures data structure inaccessible to classical kernels. Applications include molecular property prediction (classifying molecules by bioactivity using quantum-native molecular descriptors), anomaly detection in financial transactions, and materials property classification.

In PoUQW, quantum kernel evaluations performed by miners can be consumed by on-chain AI agents or off-chain consumers via oracle patterns. A QML service contract accepts data points from users, posts kernel evaluation tasks to the mining network, and returns classification or regression results computed using the quantum kernel matrix.

A concrete scenario: a DeSci protocol for drug repurposing maintains a database of molecular descriptors on-chain. Users submit candidate molecules, and the protocol posts quantum kernel evaluations as mining tasks — computing the kernel matrix between the candidate and a reference set of known-active compounds. The resulting similarity scores, verified on-chain, drive an automated recommendation engine that ranks repurposing candidates by predicted bioactivity.

7.6 Materials Science

The discovery of new materials — for batteries, catalysts, semiconductors, superconductors, and structural applications — depends critically on accurate computation of electronic structure properties: band gaps, formation energies, phonon spectra, and magnetic ordering. These properties are determined by the ground-state and excited-state solutions of crystal-structure Hamiltonians, which are quantum many-body problems.

Researchers and materials companies can post crystal-structure Hamiltonians (periodic systems modeled within the Task Registry's VQE/VQD framework using embedding methods for periodic boundary conditions) as task bounties. Miners compute ground-state properties, and the verified results contribute to an **on-chain open-access materials database** — a decentralized, immutable, and verifiable repository of quantum-computed materials properties.

A concrete scenario: a consortium of battery researchers posts Hamiltonians for 20 candidate cathode materials (lithium nickel manganese cobalt oxide variants). Over the course of a week, the mining network produces VQE-verified ground-state energies and electronic structure data for each candidate. The results, stored on-chain and openly accessible, allow any researcher to screen candidates for stability and voltage characteristics without needing their own QPU access. The consortium pays for the bounties in $Q, and the data becomes a public good — an on-chain Materials Project powered by decentralized quantum computation.

7.7 Nuclear Structure Simulation

Understanding the structure of atomic nuclei — binding energies, excited states, reaction cross-sections, and decay pathways — requires solving the nuclear many-body problem: a system of interacting protons and neutrons governed by the strong nuclear force. The nuclear Hamiltonian is strongly correlated and exhibits the same exponential scaling that makes molecular electronic structure classically intractable, making it a natural target for VQE and VQD on quantum hardware.

Dumitrescu et al. [12] demonstrated VQE computation of the deuteron binding energy on IBM quantum hardware — a landmark result showing that nuclear physics is accessible to near-term quantum devices. Subsequent work has extended to triton (^{3}H), helium-4 (^{4}He), and lithium-6 (^{6}Li) ground states using chiral effective field theory Hamiltonians mapped to qubit systems. As qubit counts scale, the computable nuclear chart expands toward heavier nuclei and reaction dynamics relevant to fusion energy, nuclear waste management, and fundamental physics.

A concrete scenario: a national laboratory consortium posts nuclear shell-model Hamiltonians for medium-mass nuclei (calcium-48, nickel-56) as VQE/VQD task bounties on PoUQW. Miners compute ground-state and excited-state energies that are classically intractable at the target accuracy. The verified results contribute to the nuclear data ecosystem used by reactor designers, fusion researchers, and astrophysicists modeling stellar nucleosynthesis — a high-value scientific output produced as a byproduct of blockchain consensus.

7.8 Catalyst Design for Clean Energy

Heterogeneous catalysis underpins the global chemical industry and is central to the clean energy transition: nitrogen fixation (Haber-Bosch process consumes $\sim 2\%$ of global energy), CO_2 reduction to fuels, water splitting for green hydrogen, and fuel cell oxygen reduction all depend on catalytic surfaces whose performance is governed by quantum-mechanical electronic structure at the active site. Designing better catalysts — cheaper, more selective, more durable — requires accurately modeling transition-state energies, adsorption geometries, and reaction barriers on metal surfaces, which involve strongly correlated d-electron systems that defeat standard DFT approximations.

Quantum computers are particularly suited to this problem because the catalytic active site (typically 10–50 atoms of the metal surface plus adsorbate) maps naturally to the active-space VQE framework: the strongly correlated region is treated quantum-mechanically while the surrounding bulk is handled by a classical embedding (e.g., DMET or QM/MM). The energy differences that determine catalytic activity — barrier heights, binding energy differences between competing pathways — are often < 0.1 eV, requiring the chemical accuracy (~ 1 kcal/mol ≈ 0.04 eV) that VQE on sufficient qubits can in principle deliver.

A concrete scenario: a green hydrogen startup posts a series of VQE task bounties for the oxygen evolution reaction (OER) on candidate iridium-free catalyst surfaces — mixed metal oxides that could replace expensive noble-metal catalysts in electrolyzers. Each bounty specifies an active-site Hamiltonian (embedded cluster of 15–30 atoms, mapped to 40–80 qubits) and a target accuracy for the OER overpotential. Miners produce verified ground-state energies for each candidate surface, and the results — openly stored on-chain — enable computational screening of catalyst candidates without proprietary QPU access. The winning catalyst design accelerates the deployment of affordable green hydrogen production.

8 Token Economics: A Store of Value for Quantum Computing

The native token of the PoUQW network, **$Q**, serves as the unit of account, medium of exchange, and store of value for the on-chain quantum compute economy. This section develops the economic model and argues that $Q functions as a **commodity money whose intrinsic backing is verified quantum computational work** — the quantum analog of how Bitcoin's value is grounded in thermodynamic energy expenditure, but redirected toward productive output.

8.1 The Commodity Thesis

The economic argument for $Q as a commodity money rests on three reinforcing properties: every token represents verified useful work, the cost of producing each token is permanently tied to the frontier of quantum computation via difficulty adjustment, and demand accrues from multiple independent sources.

Every $Q token is minted only through verified useful quantum work. Unlike tokens backed solely by staking (where value is circular — tokens are valuable because they are staked, and staked because they are valuable), every $Q token in circulation represents a **provably completed quantum computation** that produced a real, classically verified result.

The token supply is not arbitrary — it is a cryptographically auditable ledger of useful scientific and commercial output.

This is a stronger backing claim than any existing token:

- Bitcoin: each BTC represents provably expended energy on hash puzzles — real resource expenditure, but producing no useful output.
- Ethereum: each ETH in circulation was either pre-mined (genesis allocation), created as PoW block rewards (pre-merge), or created as PoS rewards (post-merge). None of these represent provably useful computation.
- Bittensor: each TAO represents AI work, but the verification is subjective (scorer-dependent), meaning the quality of the backing is uncertain.

$Q is uniquely positioned: each token represents work that is *useful* (produces a real computational result), *objectively verified* (deterministic classical verification), and *physically grounded* (requires quantum hardware to produce).

Continuously increasing difficulty creates a natural price floor tied to the cost of quantum computation. The protocol's difficulty adjustment (Section 4.4) ensures that each $Q token always represents a computation at or near the frontier of current quantum capability. As quantum hardware improves — more qubits, better fidelities, faster gate times — the protocol increases difficulty: larger problem instances, deeper circuits, tighter objective thresholds. The token never becomes "easy to mine." The marginal cost of mining one $Q is determined by the cost of QPU time required to complete a qualifying quantum work proof at the current difficulty level, plus classical compute, energy, and operational costs.

This creates a production-cost-based price floor analogous to Bitcoin's mining economics. As long as $Q trades above the marginal cost of production, rational miners are incentivized to produce useful quantum work and sell the resulting tokens. If the token price falls below production cost, miners reduce activity (QPU hardware can be redirected to non-mining quantum cloud computing), supply growth slows, and price pressure reverses. The critical difference from Bitcoin is that the marginal cost is grounded in the cost of *useful* computation rather than wasteful hash puzzles — the price floor has a dual foundation: the cost of the quantum resources consumed AND the economic value of the computation produced.

Demand accrues from multiple independent sources. $Q is demanded by several distinct constituencies:

Task bounty posters. Pharmaceutical companies, logistics firms, financial institutions, and researchers must acquire $Q to post bounties for quantum computations. As the network proves its ability to deliver useful results, demand from this constituency grows with the commercial relevance of quantum computing.

Gas fees. All EVM smart contract execution on the PoUQW chain requires $Q for gas, identical to ETH's role on Ethereum. As the DeFi, gaming, and application ecosystem on PoUQW grows, gas demand grows.

Staking. Validators must stake $Q to participate in consensus. This locks a fraction of the supply, reducing circulating supply and creating demand from parties who wish to earn staking yield.

Result access fees. Quantum computation outputs stored on-chain can be gated behind $Q-denominated access fees (via the access-controlled task bounty mechanism). Research databases, materials properties, and molecular energies may be accessible only to token holders, creating data-consumption demand.

Speculative demand. Market participants who believe that quantum computing will become increasingly commercially important — and that PoUQW will capture a share of the decentralized quantum compute market — will hold $Q as a speculative investment. While speculative demand is volatile, it provides liquidity and price discovery.

Difficulty adjustment as automatic monetary policy. $Q supply growth is constrained not by a fixed supply cap but by the continuously increasing cost of production. The difficulty adjustment algorithm (Section 4.4) ensures that as quantum hardware improves and more miners enter the network, the computational requirements for mining a block increase proportionally — larger problem instances, deeper circuits, tighter objective thresholds — maintaining the target block time and preventing a flood of cheap tokens. This is structurally different from Bitcoin's difficulty adjustment, which scales only the *number of hashes* required (a quantitative scaling). PoUQW's difficulty adjusts the *complexity of the useful problem being solved* (a qualitative scaling) — as the network matures, each $Q token represents the solution to an increasingly hard and increasingly valuable quantum computational problem. The token's backing grows stronger over time, not weaker. Unlike classical cryptocurrencies where hardware efficiency improvements (ASICs, GPU generations) temporarily make mining cheaper until difficulty catches up, PoUQW's difficulty is tied to the actual computational frontier: the problems get harder in lockstep with hardware improvements. The cost per token remains pegged to the cost of frontier-level quantum computation, providing a persistently meaningful backing.

8.2 The Quantum Computing Reserve Asset Thesis

We conclude this section with a forward-looking — and clearly speculative — argument about the long-term positioning of $Q.

As quantum computing matures into critical infrastructure for multiple industries, the economic value of quantum computation will grow. Pharmaceuticals depend on it for molecular simulation; finance for portfolio optimization and risk modeling; materials science for new battery and catalyst design; logistics for combinatorial optimization at global scale; and national security for a range of classified applications. The total addressable market for quantum computation, if the technology delivers on its promise, is measured in hundreds of billions of dollars annually [2, 24].

In this scenario, $Q — as the native token of the largest decentralized, verifiable quantum compute network — would function as a **reserve asset for the quantum economy**: a frontier-cost-backed token that provides on-chain price discovery for quantum computational resources. Just as oil serves simultaneously as a commodity, a medium of exchange for energy markets, and a macro-economic benchmark, $Q could serve as the on-chain, trustless, verifiable unit of account for quantum compute.

This thesis depends on several assumptions that are not yet validated: that quantum computing delivers real commercial value at scale (not yet proven for most applications); that PoUQW

captures a meaningful share of the quantum compute market (competitive with centralized cloud providers); and that the token economy achieves sufficient liquidity and network effects to function as a coordination mechanism. We do not present this as a certainty but as a structural possibility that motivates the protocol's design.

The honest assessment is that this thesis lives or dies with quantum computing itself. If quantum hardware fails to deliver practical advantage at commercial scale, $Q reduces to a niche token for a niche computation market. If quantum computing succeeds, PoUQW — as the first protocol to natively integrate useful quantum work into blockchain consensus — is positioned to capture substantial value. The protocol is designed to be correct in either scenario: useful from day one (producing real, if modest, quantum computations) and scaling in value alongside the technology it represents.

9 Security Analysis

This section analyzes the major attack vectors against PoUQW and the protocol's defenses. We consider five categories of adversaries: Sybil attackers, proof fabricators, randomness manipulators, majority-power adversaries, and quantum attackers targeting the chain's own cryptography.

9.1 Sybil Resistance

Attack. An adversary creates many pseudonymous identities to gain disproportionate influence over the network without proportionate resource expenditure.

Defense. PoUQW is Sybil-resistant through two complementary mechanisms:

QPU hardware scarcity. Mining requires access to real quantum processing units. Unlike classical PoW (where hash power can be acquired incrementally through ASIC purchases) or PoS (where stake can be borrowed via liquid staking derivatives), QPU access is constrained by the physical scarcity of quantum hardware. An adversary cannot create 1,000 mining identities and distribute their single QPU's compute across them in a way that produces more total useful work than a single identity — the total quantum computation is bounded by the QPU's physical throughput, regardless of how many identities submit proofs. The difficulty adjustment ensures that each proof requires a computation commensurate with the block time, so splitting a QPU across identities simply reduces the probability of any single identity producing a valid proof without increasing the aggregate probability.

Stake requirements. Validator participation requires staking $Q tokens. The cost of acquiring stake to run multiple validators is economic, not computational, providing a second layer of Sybil resistance for the consensus layer. The minimum stake per validator is set to ensure that the cost of accumulating 1/3 of the total stake (the BFT corruption threshold) is economically prohibitive.

The combination of physical (QPU) and economic (stake) Sybil resistance is strictly stronger than either alone.

9.2 Proof Fabrication

Attack. A classical adversary (or a dishonest miner with a classical computer) submits a fabricated quantum work proof — one that claims to be the result of a quantum computation but was actually produced classically, or was produced by a quantum computer but with falsified results.

Defense. The verification protocol provides multiple layers of defense:

Objective recomputation. For tasks where classical verification is feasible (QAOA cut value evaluation: always; VQE energy evaluation: for $n \leq 40$ qubits; VQLS residual: always), the verifier independently computes the objective value from the submitted solution. A fabricated proof with an incorrect objective value is rejected immediately.

Statistical consistency checks. The submitted measurement bitstrings must be statistically consistent with the claimed quantum computation. Specifically:

- The marginal distributions of measurement outcomes must match the expected distributions for the claimed ansatz and parameters (chi-squared test).
- The entropy of the measurement distribution must be consistent with quantum computation on a noisy device (too little entropy suggests classical fabrication; too much suggests random noise injection).
- Correlation patterns between measured qubits must be consistent with the entanglement structure of the ansatz circuit.

A classical adversary trying to fake measurement statistics faces a computational challenge: producing bitstrings that are simultaneously consistent with a specific quantum circuit's output distribution requires either simulating the circuit (exponentially costly for $n > 50$) or guessing statistics that happen to pass all checks (negligible probability for sufficiently many statistical tests).

Difficulty thresholds. For optimization tasks (QAOA, VQE) where the difficulty threshold is set above the best known classical approximation, a classical adversary must outperform the best classical algorithms for the problem — which, by definition of the threshold setting, is infeasible in polynomial time under standard complexity assumptions.

Complexity-theoretic arguments by task class:

- **VQE.** Computing ground-state energies of local Hamiltonians is QMA-complete [20]. While this is a worst-case result and typical instances may be easier, strongly correlated molecular Hamiltonians used in the Task Registry are selected specifically to be in the classically hard regime.
- **QAOA.** For random 3-regular graph MaxCut instances at depth $p \geq 11$, numerical evidence indicates that QAOA exceeds the Goemans-Williamson bound $\alpha_{GW} \approx 0.878$ [35]. Setting $C_{\text{thresh}} > \alpha_{GW} \cdot C_{\max}$ (strictly above the GW bound) ensures that no polynomial-time classical algorithm can guarantee meeting the threshold; under the Unique Games Conjecture, approximating MaxCut beyond $\alpha_{GW} \approx 0.878$ is UGC-hard.

- **Random circuit sampling.** The random circuit sampling conjecture [1] asserts that no polynomial-time classical algorithm can produce samples with $F_{\text{XEB}} \geq f_{\text{thresh}}$, where f_{thresh} is a noise-adjusted protocol parameter set below the ideal value of 1 but above the classical baseline of 0. For NISQ devices operating near the quantum-classical boundary, typical targets are $f_{\text{thresh}} \in [0.1, 0.3]$ depending on circuit depth and qubit count. While this conjecture is debated ([27] demonstrated efficient classical simulation of certain circuit geometries), the difficulty parameters can be adjusted to target circuit sizes and depths that remain beyond practical classical simulation.

9.3 Randomness Manipulation

Attack. An adversary attempts to bias the beacon randomness to influence validator selection in their favor.

Defense. The SC-QRNG beacon provides multiple layers of defense:

Quantum-sourced entropy with live certification. Each beacon node's SC-QRNG produces randomness from a physical quantum process (vacuum fluctuations, photon detection, or laser phase noise) whose min-entropy is continuously certified. Unlike a classical PRNG, the entropy source cannot be replayed, predicted, or controlled without physically tampering with the trusted measurement part of the device (in a way that would still be detectable by periodic tomographic monitoring).

XOR aggregation. The beacon output is the XOR of all contributing beacon nodes' outputs. This ensures that the aggregate output is unbiased as long as one beacon opens its committed output of certified randomness. An adversary controlling $B - 1$ of B beacon nodes still cannot bias the output.

Commit-reveal. The commit-reveal structure prevents adaptive attacks: each beacon node commits to its output before seeing others' outputs, eliminating the possibility of beacons repeatedly sampling randomness to achieve a favorable result.

Multi-vendor diversity. The protocol requires beacon nodes to use SC-QRNGs from at least two distinct manufacturers, so a supply-chain attack on a single vendor's firmware cannot compromise the beacon.

Cost of attack. To bias the beacon, an adversary would need to simultaneously: (a) control all B beacon nodes, (b) compromise every node's SC-QRNG hardware (requiring independent firmware attacks on devices from multiple vendors), AND (c) produce correlated outputs that achieve a desired bias while maintaining valid entropy certificates. The multi-vendor requirement and independent tomographic monitoring make this a qualitatively harder attack than biasing classical RANDAO, where a single validator can simply choose a favorable value. The security assumption is that at least one of B beacon nodes operates an uncompromised SC-QRNG — a weaker assumption than RANDAO's requirement for economic rationality across all participants.

9.4 Majority Power Attack

Attack. An adversary acquires majority quantum computational power (the quantum analog of a 51% attack in Bitcoin) or majority stake (the standard BFT attack).

Defense. PoUQW's dual-resource model requires the adversary to achieve majority on *both* dimensions:

Quantum work majority. To consistently produce block-winning work proofs, the adversary needs more quantum computational throughput (QPU time $\times$ qubit count $\times$ gate fidelity) than the honest network. Given the physical scarcity of QPU hardware and the capital costs involved ($10M+ per high-end QPU system), acquiring majority quantum compute is vastly more expensive than acquiring majority hash power in Bitcoin (where ASICs are commoditized) or majority GPU power in Bittensor.

Stake majority. To subvert BFT finality, the adversary needs $\geq 1/3$ of the total stake. This is a standard PoS security assumption, defended by the economic cost of acquiring and risking that much stake.

Combined attack threshold. The most dangerous attack — chain reorganization with finality reversal — requires both majority useful work (to produce a longer alternative chain) AND $\geq 1/3$ of stake (to prevent the honest chain from finalizing). Either condition alone is insufficient: majority work without stake influence produces blocks that honest validators finalize on the canonical chain; majority stake without work cannot produce valid blocks (which require genuine quantum work proofs).

This dual-resource requirement significantly increases the cost of a successful attack compared to single-resource systems.

9.5 Quantum Threats to Cryptography

Attack. An adversary with a sufficiently powerful quantum computer attacks the cryptographic primitives used by the PoUQW protocol itself — digital signatures, hash functions, and commitments.

Defense. PoUQW uses **post-quantum cryptography** throughout the protocol stack:

Digital signatures. All signatures (miner signatures on work proofs, validator signatures on attestations, beacon node signatures on contributions) use **CRYSTALS-Dilithium** [11], a lattice-based signature scheme standardized by NIST (FIPS 204) and believed to be secure against quantum adversaries. Dilithium's security is based on the hardness of the Module Learning With Errors (MLWE) problem, for which no polynomial-time quantum algorithm is known.

Hash functions. The protocol uses **SHA-3 (Keccak)** for hashing (state roots, transaction hashes, commitment schemes). SHA-3 is quantum-resistant: Grover's algorithm provides a quadratic speedup for hash preimage search ($O(2^{n/2})$ instead of $O(2^n)$), but with a 256-bit hash, this still requires $O(2^{128})$ quantum operations — well beyond the capability of any foreseeable quantum computer.

Key exchange / encryption. Where encryption is used (e.g., access-controlled task bounty results), the protocol uses **CRYSTALS-Kyber** (FIPS 203), a lattice-based key encapsulation mechanism, also standardized by NIST.

Address derivation. Unlike Ethereum (which uses ECDSA and is vulnerable to quantum attacks on the secp256k1 curve), PoUQW addresses are derived from Dilithium public keys, making them quantum-safe from genesis.

This makes PoUQW an inherently post-quantum blockchain: it *uses* quantum hardware for consensus while being *resistant* to quantum attacks on its own cryptographic infrastructure. This is a distinctive and important design property — the chain benefits from quantum computing while being immune to its cryptographic threats.

10 The PoUQW Backbone Protocol

This section formalizes the round-based protocol that every PoUQW node executes. We present two algorithms: the per-slot node protocol (Algorithm 1) and the classical verification of quantum work proofs (Algorithm 2).

10.1 Notation

- $\mathcal{C}$: the node's local finalized chain.
- $\mathcal{T}(\mathcal{C})$: on-chain Task Registry at head of $\mathcal{C}$ (Section 4.1).
- R_e: beacon randomness for epoch e (Section 5.2).
- $\textsc{proposer}(t)$: beacon-derived proposer for slot t via $\textsc{StakeWeightedSelect}(\textsc{Hash}(R_e \| t \| \texttt{"proposer"})$, stake_table) (Section 5.3.2).
- $\textsc{committee}(t)$: beacon-derived attestor committee of size $C = 128$ for slot t.
- E_{thresh}: difficulty-adjusted energy threshold for minimization tasks such as VQE (Section 4.3). Acts as a ceiling on the ground-state energy estimate — a miner's solution E^* must satisfy $E^* \leq E_{\text{thresh}}$ to constitute valid work. The threshold tightens or relaxes via the dual-timescale difficulty adjustment to maintain the target 12-second slot cadence.
- C_{thresh}: difficulty-adjusted cut-value threshold for maximization tasks such as QAOA (Section 4.3). Acts as a floor on the objective value — a miner's best cut C^* must satisfy $C^* \geq C_{\text{thresh}}$ to constitute valid work. Adjusted by the same dual-timescale mechanism as E_{thresh}.

10.2 Algorithm 1: Per-Slot Node Protocol

Each slot t lasts 12 seconds. The beacon output R_e deterministically fixes the proposer and committee schedule for all slots in epoch e (Section 5.3.2). The quantum work function $Q(\cdot)$ dispatches to the appropriate subroutine (Algorithm 3 or 4) based on the task class returned by $\textsc{SelectTask}$.

Algorithm 1 PoUQW Per-Slot Node Protocol (slot t, epoch e)

— *Phase 1: Receive and validate* —
1: Receive pending transactions and any proposed block B_t from the P2P network
2: if B_t received **then** validate B_t using Algorithm 2
— *Phase 2: Propose (if selected)* —
3: if my_address = PROPOSER(t) **then**
4: task ← SELECTTASK$\big(\mathcal{T}(\mathcal{C}),\ R_e,\ t\big)$
5: v_{thresh} ← GETTHRESHOLD(task, $\mathcal{C}$)
6: (sol, v^*, Z, meta) ← Q(task, qpu)
7: **if** MEETSTHRESHOLD(v^*, v_{thresh}, task.class) **then**
8: π ← BUILDPROOF(task, sol, v^*, Z, meta)
9: B_t ← ASSEMBLEBLOCK($\mathcal{C}$, txns, π, t, σ) ▷ *sign with Dilithium*
10: DIFFUSE(B_t)
11: **end if**
12end if
— *Phase 3: Attest (if on committee)* —
13if my_address ∈ COMMITTEE(t) **and** B_t is valid **then**
14: DIFFUSE$\big($SIGNATTESTATION(B_t)$\big)$
15end if
— *Phase 4: Finalize* —
16if $\geq 2f{+}1$ attestations received for B_t **then**
17: $\mathcal{C} \leftarrow \mathcal{C} \| B_t$ ▷ *append finalized block*
18end if

The threshold test at line 7 is task-class-dependent: for minimization tasks (VQE, VQD, VQLS) it checks $v^* \leq v_{\text{thresh}}$; for maximization tasks (QAOA) it checks $v^* \geq v_{\text{thresh}}$. A block is valid if and only if it contains (Section 6.1): (1) a `QuantumWorkProof` passing Algorithm 2 under current difficulty; (2) a valid transaction set; (3) a proposer address matching PROPOSER(t); and (4) consistent epoch and beacon references. The block reward is split 70% to the QPU miner who produced the work proof and 30% to the proposer (Section 6.4.1).

10.3 Algorithm 2: Classical Verification

Invoked by every node on each received block. Runs entirely on classical hardware (Section 3.4). The verification procedure dispatches on the task class; we present the general structure below. Task-class-specific verification details are given in Section 4.1.

Algorithm 2 Classical Verification of a PoUQW Block

Input: block B with `QuantumWorkProof` π **Output:** VALID / INVALID

— *Parse and authenticate* —

1: (task_id, sol, v_{claimed}, Z, M, meta, σ) $\leftarrow \pi$

2: Verify CRYSTALS-Dilithium signature σ ▷ *reject if invalid*

— *Deterministic objective evaluation (Section 4.1)* —

3: task $\leftarrow$ FETCHTASK(task_id, parent(B))

4: $v_{\text{ver}} \leftarrow$ EVALOBJECTIVE(task, sol) ▷ *classical recomputation*

5: if $|v_{\text{ver}} - v_{\text{claimed}}| > \varepsilon_{\text{tol}}$ **then return** INVALID

— *Threshold check (Section 4.3)* —

6: if not MEETSTHRESHOLD(v_{ver}, v_{thresh}, task.class) **then return** INVALID

— *Statistical anti-spoofing (Section 9.2)* —

7: if not STATISTICALCHECK(task, sol, Z, M) **then return** INVALID

— *Physical plausibility* —

8: if v_{ver} outside physical bounds **or** meta.fidelity $< F_{\min}$ **then return** INVALID

9: return VALID

For VQE/VQD tasks, EVALOBJECTIVE reconstructs the Hamiltonian $H = \sum_i c_i P_i$ and computes $E_{\text{ver}} = \sum_i c_i \langle P_i \rangle_{\theta^*}$ at cost $O(N_P \cdot \text{poly}(n))$, with $N_P = O(n^4)$ Pauli strings. For QAOA tasks, EVALOBJECTIVE evaluates the classical cost function $C(z^*) = \sum_{(i,j)\in E} w_{ij}(1 - z_i^* z_j^*)/2$ at cost $O(|E|)$. The STATISTICALCHECK performs χ^2 tests comparing submitted measurement bitstrings against Born-rule expectations (VQE/VQD) or verifies the consistency of the reported bitstring distribution with the QAOA circuit structure (QAOA). See Sections 4.1.1 and 4.1.2 for full details.

10.4 Protocol Lifecycle

A new node joining the PoUQW network executes the following sequence:

1. **Genesis and chain sync.** Initialize with the hardcoded genesis block and sync the full chain via P2P, classically verifying every block's quantum work proof using Algorithm 2. No quantum hardware is needed for sync.

2. **Role activation.** The operator selects one or more roles (Section 3.4): *full node* (default, no staking or QPU required), *validator* (stakes `$Q`, eligible for beacon-derived committee selection), *miner* (registers QPU, produces quantum work proofs via Algorithms 3/4), or *beacon node* (registers SC-QRNG, contributes to the randomness beacon).

3. **Steady-state operation.** Each 12-second slot, the node executes Algorithm 1: receive and validate any proposed block, propose if beacon-selected for this slot (assembling a block with a valid quantum work proof), attest if on the committee, and finalize once a supermajority of attestations is reached.

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